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Review

A Comprehensive Review of Compact Multi-Port mmWave MIMO Antenna Systems for 5G/6G: Performance, Materials, and Smart Integration

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Abstract

The rapid evolution of fifth-generation (5G) and emerging sixth-generation (6G) wireless communication systems has considerably intensified the need for high data rates, ultra-low latency, massive connectivity, and intelligent network integration. To satisfy these requirements, millimeter-wave (mmWave) bands offer large available bandwidths; however, their severe propagation losses and integration constraints necessitate advanced antenna solutions. In this context, compact multi-port Multiple-Input–Multiple-Output (MIMO) antennas are a key solution for high-capacity and reliable mmWave communications. This review presents a comprehensive overview of recent antenna system technologies for 5G/6G applications, focusing on small mmWave MIMO antenna designs, performance improvement methods, advanced materials, and smart integration methods. Several antenna structures, such as microstrip patch, dielectric resonator, slot-based, and metamaterial-inspired designs, are critically discussed and compared. In addition, this review analyzes key design challenges involving miniaturization, mutual coupling reduction, bandwidth enhancement, gain improvement, radiation efficiency, and integration complexity, along with their impact on key performance metrics. The importance of advanced materials, artificial-intelligence-assisted optimization, hybrid antenna architectures, and smart integration strategies in future 5G/6G systems is also emphasized. Finally, we identified current challenges, emerging trends, and future research directions to provide useful design guidelines for researchers and engineers developing next-generation high-performance antenna systems for intelligent wireless communications.

Keywords: mmWave antennas; MIMO systems; 5G/6G communications; mutual coupling reduction; compact antenna design; metamaterials; diversity performance

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1. Introduction

In the current fast-changing digital landscape, there is an increasing need for significantly higher data rates, ultra-reliable low-latency communication, and support for a vast number of connected devices. Fifth-generation (5G) and advanced sixth-generation (6G) mobile networks are being developed to address these demands. Reaching these

performance levels requires tapping into new frequency bands, most notably the millimeter-wave (mmWave) spectrum beyond the already crowded sub-6 GHz range [1], as depicted in Figure 1.

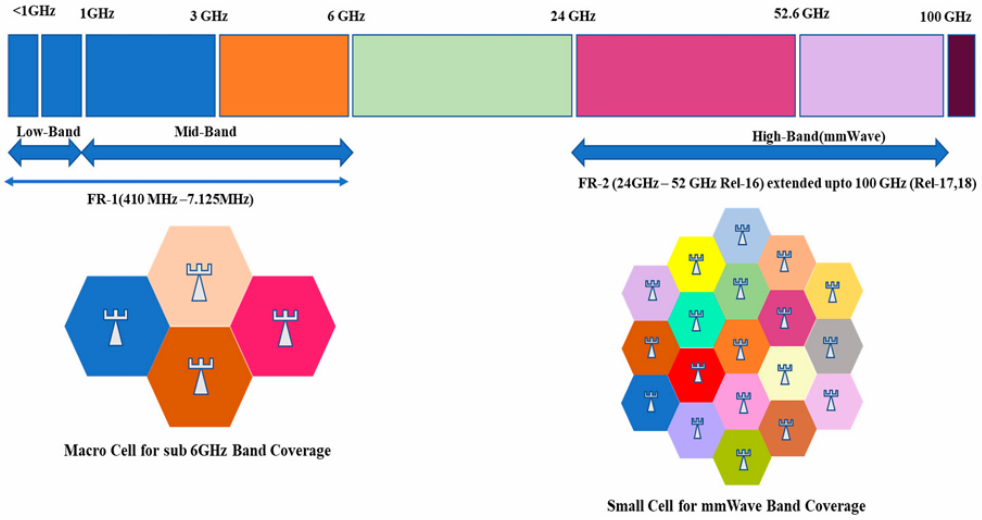


Figure 1. Frequency spectrum allocation highlighting sub-6 GHz and millimeter-wave bands, adapted from [2].

These frequency bands, typically ranging from 24 GHz to 100 GHz, offer large contiguous bandwidths [3]. In particular, the bands around 28 GHz, 38 GHz, and 60 GHz have been extensively studied, enabling multi-gigabit-per-second data rates and enhanced system capacity [4]. However, substantial free-space propagation losses, high sensitivity to blockage, atmospheric attenuation, and restricted communication range continue to considerably limit the effective use of mmWave frequencies. In this context, Multiple-Input–Multiple-Output (MIMO) systems integrated with beamforming techniques are essential elements of contemporary 5G/6G architectures, facilitating significant enhancements in channel capacity, link reliability, and radiation gain [5]. Nevertheless, the integration of compact multi-port mmWave MIMO antennas into space-constrained platforms continues to represent a major engineering challenge due to the complex trade-offs among antenna miniaturization, inter-element isolation, impedance bandwidth, radiation efficiency, and fabrication complexity [6].

Despite significant progress in mmWave MIMO antenna design, achieving an optimal balance between compactness and electromagnetic performance remains a critical challenge. Reducing the spacing between antenna elements in compact platforms generally increases mutual coupling, also fabrication tolerances, and dielectric and conductor losses [7,8], which degrades key MIMO performance metrics such as isolation, envelope correlation coefficient (ECC), diversity gain (DG), and channel capacity [9].

To overcome these limitations, a wide range of decoupling and performance-enhancement techniques has been investigated, including defected ground structures (DGSs), electromagnetic bandgap (EBG) structures, parasitic elements, polarization diversity, metasurfaces, and metamaterial-based approaches. Although these techniques can significantly improve isolation and bandwidth characteristics, they often introduce additional design complexity, fabrication sensitivity, increased cost, or reduced scalability for high-density multi-port configurations.

The objective of this review is to present a comparative analysis and furnish a thorough examination of compact multi-port MIMO antenna designs for millimeter-wave 5G and beyond wireless communications. The article focuses on how various antenna topologies,

design methods, and substrate technologies impact key performance metrics, including gain, isolation, bandwidth, and diversity.

The review's primary objective is to address the following questions: What are some ways that improved antenna systems are used in 5G and the new 6G wireless communications? What are the benefits of these antenna designs in terms of channel capacity, spectral efficiency, dependability, and the ability to work with other systems? What are their inherent limitations and challenges? Furthermore, finally, what emerging trends and future directions are shaping the development of next-generation MIMO antennas? By addressing these questions, this review seeks to provide clear guidance for researchers and designers.

This work specifically seeks to attain three major targets. It starts by identifying and categorizing the main design strategies applied in compact MIMO antenna systems, Second, it compares the electrical performance and potential for physical integration of different designs, considering practical factors for substrate selection and device integration. Third, by highlighting new developments in antenna topologies, manufacturing techniques, and innovative substrate materials that can direct the creation of next-generation mmWave MIMO antennas, the review aims to highlight emerging research trends and future directions.

Recent research on mmWave and beyond-mmWave antennas for 5G/6G shows a clear push toward compact multi-port MIMO structures that can maintain wide bandwidth, high gain, and strong port isolation despite the severe propagation loss and tight integration constraints at high frequencies. For instance, surveys by [10,11] highlight that many mmWave antennas still suffer from bandwidth and radiation efficiency limitations. As a result, modern designs increasingly rely on antenna arrays, multi-port MIMO configurations, and electromagnetic manipulation techniques such as slots, stubs, and metamaterial/metasurface-inspired structures to improve gain, impedance matching, and isolation. In the same direction, recent works such as [12,13] discuss advanced integration trends, including antenna-in-package (AiP), antenna-on-chip (AoC), low-loss substrates, and emerging fabrication technologies. These studies emphasize that future mmWave antennas must be designed to comply with practical packaging, thermal, and system-level integration constraints. Moreover, mutual coupling mitigation has been consistently identified as one of the main bottlenecks in compact MIMO systems. Comprehensive reviews such as [2,14] summarize various decoupling strategies, including EBG/FSS structures, neutralization lines, parasitic decoupling elements, and metamaterials.

However, while earlier reviews looked at the problems with 5G mmWave, more current ones are more focused on 6G antenna technologies: For instance, refs. [15,16] covering frequency bands from mmWave to sub-THz, miniaturization techniques, distributed MIMO architectures, material selection, packaging strategies, and radiation performance requirements toward future 6G deployment. Emerging surveys discuss key enabling technologies in the context of MIMO advancements, such as massive/XL MIMO, intelligent reflecting surfaces, and other spatial multiplexing techniques envisioned for 6G networks that extend the scope of compact multi-antenna systems [17]. By concentrating on these goals, the review offers researchers and designers working on high-frequency MIMO systems a clear framework for comprehending current applications, assessing performance trade-offs, and identifying unresolved issues in the field.

Overall, Table 1 shows that existing review articles have addressed several important aspects of mmWave, MIMO, and 6G antenna technologies, but most of them treat these topics separately. Some works mainly focus on general mmWave antenna designs, while others emphasize MIMO antenna structures, mutual coupling reduction, fabrication methods, AI-assisted optimization, or 6G-oriented antenna architectures. Therefore, a unified review specifically dedicated to compact multi-port mmWave MIMO antennas, while

jointly considering isolation, fabrication, AI/ML optimization, CMA, and 6G scalability, remains limited.

Table 1. Summary of recent review papers on mmWave, MIMO, and 6G antenna technologies.

Ref.	Year	Main Focus	Compact Multi-Port MIMO	Isolation Techniques	Fabrication and Integration	AI/Optimization	6G Perspective	Main Limitations
[2]	2023	5G MIMO antenna designs and decoupling techniques	Yes	Extensive	Partial	No	Partial	Limited focus on AI, fabrication, and 6G/THz integration.
[14]	2023	5G MIMO antennas and mutual coupling reduction	Yes	Extensive	Partial	No	Partial	Limited focus on compact mmWave MIMO and 6G-oriented designs.
[17]	2023	Emerging MIMO technologies for 6G networks	Partial	Limited	No	Partial	Extensive	Limited discussion of antenna geometry, isolation, and fabrication.
[10]	2024	General mmWave antenna survey	Partial	Limited	Extensive	No	Partial	Limited focus on compact MIMO, isolation, and AI optimization.
[11]	2024	mmWave MIMO antenna structures and isolation techniques	Yes	Extensive	Partial	No	Limited	Limited 6G, AI, and fabrication-aware analysis.
[16]	2024	6G antenna architectures from mmWave to THz bands	Partial	Limited	Extensive	Partial	Extensive	Limited analysis of compact MIMO isolation and experimental validation.
[12]	2025	5G mmWave antennas, MIMO, beamforming, and integration	Partial	Moderate	Extensive	Partial	Extensive	Limited focus on compact multi-port MIMO topologies.
[13]	2025	mmWave antennas, arrays, beam-steering, and emerging technologies	Partial	Moderate	Extensive	Extensive	Extensive	Limited analysis of dense multi-port MIMO isolation.
[15]	2026	6G antenna technologies, THz, RIS, materials, and packaging	Partial	Moderate	Extensive	Extensive	Extensive	Limited focus on compact mmWave multi-port MIMO isolation.
This work	2026	Compact multi-port mmWave MIMO antennas for 5G/6G	Extensive	Extensive	Extensive	Extensive	Extensive	Unified review addressing compactness, isolation, fabrication, AI optimization, and 6G scalability.

In contrast, the present review provides a focused and integrated analysis of compact multi-port mmWave MIMO antennas for 5G, 5G-advanced, and future 6G communication systems. Its contribution lies in jointly examining antenna compactness, isolation performance, bandwidth and gain enhancement, radiation efficiency, diversity metrics such as ECC, DG, TARC, and CCL, substrate and fabrication constraints, AI/ML-based optimization, characteristic mode analysis, and 6G scalability. Therefore, this work differs from previous reviews by connecting electromagnetic antenna design, MIMO performance evaluation, intelligent optimization methods, and practical deployment constraints within a single review framework.

Moreover, this review places particular emphasis on implementation-oriented challenges, including feeding network complexity, packaging constraints, fabrication tolerance, material losses, thermal effects, RF front-end integration, and manufacturability. It also discusses future research directions related to reconfigurable antennas, smart metasurfaces, sub-THz/THz operation, IoT, V2X, self-powered antennas, and eco-friendly materials. This unified perspective provides researchers and antenna designers with a clearer framework for identifying current design trade-offs, unresolved technical bottlenecks, and future development paths for compact multi-port mmWave MIMO antenna systems.

To address the objectives of this review, the paper is organized as follows:

- Summarizes the key contributions, including the proposed taxonomy, main performance metrics, and design insights for compact mmWave MIMO antennas.

- Introduces the principles of mmWave communications, MIMO theory, presents the key performance metrics for MIMO systems (ECC, DG, TARC, CCL), and antenna structures, including single-element performance metrics.
- Reviews techniques for mutual coupling reduction, performance optimization, and advanced AI/ML-based design methodologies. Furthermore, presents representative multi-port MIMO antennas from the literature.
- Discusses electromagnetic and physical constraints, substrate characteristics, compares common and emerging materials, highlights design limitations, and integration and manufacturing Challenges.
- Outlines current trends and future research directions, including reconfigurable antennas and integration with 6G/IoT applications.
- Summarizes key findings, emphasizes the research's impact, and offers final perspectives on the field.

2. Contributions of This Paper

To provide a structured overview of compact MIMO antenna technologies for 5G/6G mmWave communications, a comprehensive taxonomy is proposed in Figure 2. This taxonomy is used as an organizing framework for the review and connects the main parts of the manuscript. It organizes the literature into five main groups: the fundamentals of mmWave MIMO systems, antenna structures, design methods, performance evaluation metrics, current challenges, and future research directions. The fundamentals and antenna-structure branches are mainly discussed in Section 3, while the design-method branch is developed in Section 4 through mutual coupling reduction approaches, structural and feeding-based optimization techniques, and advanced design methodologies. The performance-evaluation branch supports the comparative analysis of key metrics such as gain, bandwidth, isolation, ECC, DG, TARC, and CCL, whereas the challenges and future-direction branches are addressed in Sections 5 and 6.

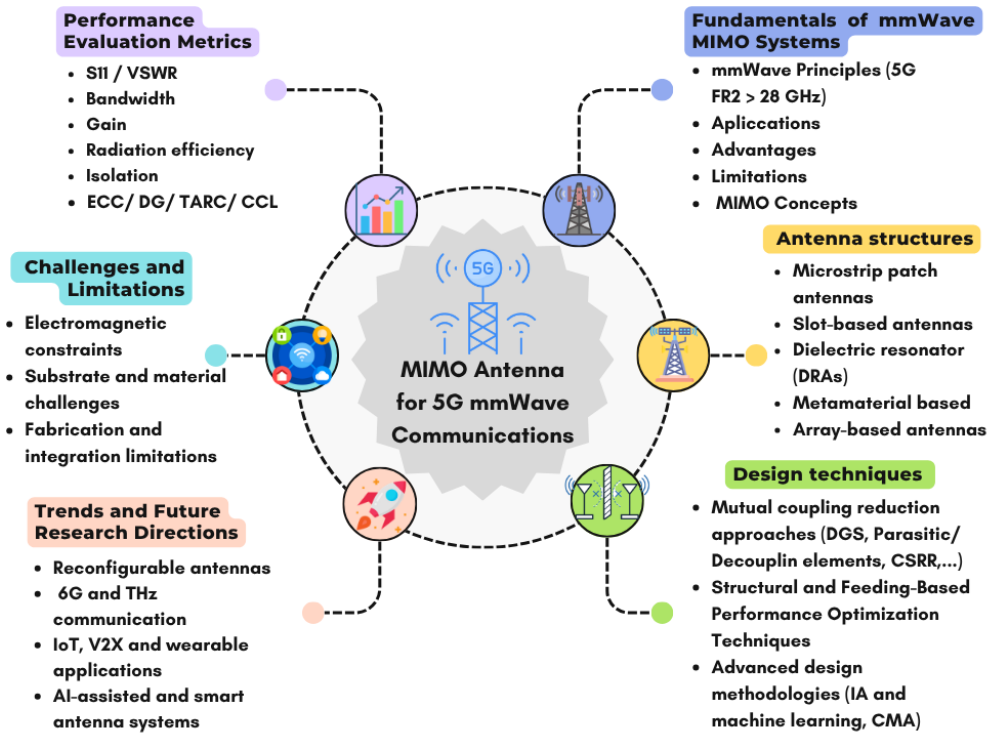


Figure 2. Taxonomy of compact mmWave MIMO antenna research for 5G and beyond.

2.1. Key Performance Metrics

Beyond conventional parameters, MIMO performance metrics offer a common basis for evaluating and comparing compact mmWave antennas In practical design, a high-performance MIMO antenna is anticipated to show a low ECC, typically below 0.5 and ideally below 0.05 or 0.02, indicating minimal correlation between radiating elements. Similarly, DG should be close to 10 dB, reflecting efficient diversity performance [18].

In addition, the total active reflection coefficient (TARC) is used to evaluate the overall impedance matching of the multi-port system under simultaneous excitation, where lower values indicate better performance (less than 0 dB) [18]. The channel capacity loss (CCL), on the other hand, quantifies the degradation in channel capacity caused by mutual coupling and correlation effects and should commonly remain below 0.4 bits/s/Hz for efficient MIMO operation [9,19].

Therefore, the joint analysis of ECC, DG, TARC, and CCL enables a more reliable comparison of compact mmWave MIMO antennas, since a design with good isolation alone may still suffer from suboptimal diversity or capacity performance.

To highlight the performance of recent compact mmWave MIMO antenna designs, a comparative study is presented in Table 2. The comparison focuses on the main MIMO performance metrics, including operating frequency, antenna size, isolation, gain, and our metrics (ECC, DG, TARC, CCL). These parameters are essential for evaluating the suitability of MIMO antennas for 5G and beyond mmWave communication systems, since they directly reflect the antenna compactness, diversity behavior, mutual coupling reduction, and channel performance.

Table 2. Comparative analysis of recent compact mmWave MIMO antennas based on key MIMO performance metrics.

Ref.	Data Type	Frequency (GHz)	Size (mm^3)	Isolation (dB)	Gain (dBi)	ECC	DG (dB)	TARC (dB)	CCL (bits/s/Hz)	Technique Used
[20]	Measured/ Simulated	26.36–41.57	23 × 17 × 0.254	>23.2	6.2	0.0008	9.9998/9.9999	<−10	0.08/0.13	Triple-triangle rings with trapezoidal slot and connected ground plane
[21]	Measured/ Simulated	25–38	20 × 20 × 0.254	>18	6.2	<0.012	≈10	<−10	<0.5	Plow-shaped radiators with orthogonal arrangement for polarization diversity
[22]	Measured/ Simulated	3.59–3.85/ 8.40–70.0	70.71 × 70.71 × 0.254	>15	16.13	<0.10	>9.9998	<−8	<0.10	Eight-port flexible MIMO with oval slotted patches, rectangular DGS, asymmetric feed and central decoupling structure
[23]	Measured/ Simulated	24.6–31.8	15 × 15 × 0.5	>22.5	8.1	<0.007	≈10	<−10	<0.25	Hybrid decoupling using orthogonal antenna orientation and defected ground structure (DGS)
[24]	Measured/ Simulated	24–40	35 × 35 × 0.254	>44	18.2	<0.0005	9.99	<0	<0.3	Quad-port MIMO array using diamond-shaped slots and defected ground geometry (DGG)
[25]	Measured/ Simulated	28/38	32 × 32 × 0.787	>21/>20	6.4/8.3	0.0001/ 0.00007	≈10	<−10	0.06/0.07	Curved-edge radiating patches with partial ground planes connected by 45° tilted strips
[26]	Measured/ Simulated	25–29.5	13 × 13 × 0.8	>25	10	0.0003	≈10	≈−10	0.03	Inverted L-shaped radiators with hexagonal slotted DGS and orthogonal quad-port placement

Based on Table 2, the most recent compact mmWave MIMO antennas demonstrate good diversity performance, particularly in terms of ECC and DG. Several works, such as [24–26], achieve extremely low ECC values below 0.001, which indicates a very weak correlation between the MIMO elements and confirms efficient diversity behavior. Other designs, including [20,21,23], also maintain ECC values below the acceptable MIMO threshold,

showing that low correlation has become a common design objective in recent mmWave antenna systems. However, the differences between the works become more visible when considering TARC and CCL. The antennas in [20,21,23,25,26] maintain TARC values below −10 dB, indicating good impedance matching when multiple ports are excited simultaneously. This trend shows that recent mmWave MIMO designs generally satisfy the basic diversity requirements for 5G and beyond wireless systems. In contrast, ref. [24] reports TARC values below 0 dB only, which is acceptable but less strict compared with the designs reporting values below −10 dB.

Regarding channel capacity loss, most antennas exhibit CCL values below the standard acceptable limit. The lowest CCL values are obtained in [20,25,26], these results indicate better channel utilization and reduced capacity degradation. Meanwhile, designs such as [21,23] remain within acceptable limits but present relatively higher CCL values, mainly due to their wider bandwidth, flexible structure, or compact integration constraints.

Overall, the comparison shows a clear trend toward compact mmWave MIMO antennas. The main trade-off observed is between antenna compactness, operating bandwidth, gain, and isolation enhancement technique. Designs using defected ground structures, orthogonal element placement, and slot-based decoupling generally achieve better isolation and diversity performance. Therefore, the proposed comparison highlights that an efficient mmWave MIMO antenna should not only provide high gain and wide bandwidth, but also maintain low correlation, low channel capacity loss, and stable multi-port matching.

2.2. Analysis of Design Trade-Offs in Compact mmWave MIMO Antennas

Compact multi-port MIMO antennas for mmWave applications involve several critical design trade-offs, where improving one performance metric often leads to the degradation of another. These trade-offs are essential for optimizing antenna performance in next-generation wireless systems.

One of the key design trade-offs in compact mmWave MIMO antennas lies in the relationship between element size and mutual coupling. Reducing antenna dimensions generally leads to closer spacing between radiating elements, which increases electromagnetic interaction and degrades isolation performance. For instance, a 4×4 antenna array reported by Edries et al. achieves high isolation (>26 dB) and extremely low ECC (<0.001) through optimized geometry and slot implementation. However, this performance is obtained at the expense of increased design complexity and a moderate footprint of approximately $40 \times 40 \times 0.8$ mm^3, clearly illustrating the trade-off between compactness and isolation performance [27].

In contrast, a highly compact dual-band MIMO antenna occupying only 15×15 mm^2 demonstrated excellent MIMO metrics (ECC $\approx$ 0.001, diversity gain $\approx$ 9.99 dB, and isolation $>$ 25 dB), indicating that careful electromagnetic design can partially mitigate coupling effects even in reduced-size configurations [28]. Nevertheless, achieving such performance in compact structures often requires precise optimization and may not be easily scalable to higher-order MIMO systems.

Another important trade-off exists between isolation enhancement techniques and other antenna performance parameters. For example, incorporating defected substrate structures in a quad-port mmWave MIMO antenna improved isolation from approximately 17 dB to 24 dB while simultaneously increasing gain by 3–4 dB. Although this demonstrates the effectiveness of decoupling structures, it also suggests that such modifications can significantly alter current distribution, thereby influencing bandwidth and radiation characteristics [29].

Similarly, wideband eight-port MIMO designs have been reported to achieve high isolation (around 30 dB) and excellent diversity performance (ECC $<$ 0.0001) over the

28–46 GHz range. However, these designs typically require larger physical dimensions, highlighting the trade-off between bandwidth, isolation, and overall device footprint [30]. These observations confirm that enhancing one performance metric often impacts others, making simultaneous optimization particularly challenging.

Finally, advanced optimization approaches, such as machine-learning-based antenna design, have emerged as promising tools to efficiently explore complex trade-offs between size, gain, bandwidth, and isolation. These techniques enable multi-objective optimization and may provide more balanced design solutions compared to conventional methods, thereby opening new perspectives for compact mmWave MIMO antenna development [31].

2.3. Research Gap Identification

Recent developments in compact MIMO antennas for millimeter-wave (mmWave) 5G and beyond communication systems have demonstrated significant progress in terms of bandwidth enhancement, high gain, mutual coupling reduction, and multi-port integration [13]. However, despite these advances, several research gaps remain open and require further investigation.

One of the most important challenges is the trade-off between compactness, high gain, wide bandwidth, and high isolation. Although recent works have proposed compact dual-band and wideband MIMO antennas with promising performance, achieving all these requirements simultaneously remains difficult. For example, Basha et al. [32] proposed a compact dual-band four-port MIMO antenna operating at 28/38 GHz with high gain and good isolation. Similarly, Tiwari et al. [24] presented a wideband quad-port MIMO antenna with strong isolation and high gain for 5G NR mmWave applications. Nevertheless, these designs still show that increasing gain and bandwidth often requires additional structural modifications such as slots, defected ground structures, corporate-feed networks, or array configurations, which may increase the design complexity and affect the overall compactness of the antenna.

Another major research gap concerns mutual coupling reduction in compact MIMO configurations. In MIMO systems, closely spaced antenna elements can lead to strong electromagnetic interaction, which degrades key performance metrics. Kaur et al. [33] reviewed several isolation techniques for 5G MIMO antennas, including approaches used for sub-6 GHz, 28 GHz, and 28/38 GHz bands. However, many existing decoupling techniques improve isolation at the cost of increased antenna size, fabrication complexity, or narrowband behavior. Therefore, there is still a need for simple, low-profile, and fabrication-friendly isolation techniques that can maintain high performance without compromising compactness.

The integration of a larger number of antenna ports is another open issue for future 5G and beyond systems. These are essential to improving channel capacity, diversity performance, and spatial multiplexing. Khan et al. [30] proposed a wideband eight-port MIMO antenna with reduced mutual coupling for future 5G mmWave applications. Although this type of architecture is promising, increasing the number of ports generally leads to stronger coupling, more complex feeding networks, higher design sensitivity, and larger physical dimensions. Thus, future research should focus on scalable MIMO architectures that can integrate four, eight, or more ports while preserving compact size, high isolation, stable radiation patterns, and low ECC.

In addition, most reported compact mmWave MIMO antennas are designed for fixed frequency bands, especially 28 GHz and 38 GHz. Mehmood et al. [13] highlighted the importance of reconfigurable antennas, beam-steering techniques, phased arrays, metasurfaces, and advanced materials for future mmWave and 6G systems. However, many compact MIMO designs still lack frequency reconfigurability, polarization reconfigurability,

or beam-steering capability. This limitation reduces their adaptability to future communication scenarios, where user mobility, blockage, and dynamic channel conditions are critical issues.

Recently, machine learning has emerged as a promising approach to reduce the time-consuming process of electromagnetic simulation and antenna optimization. Haque et al. [9] proposed a machine-learning-based technique for gain prediction of a compact 28 GHz MIMO antenna and demonstrated that data-driven models can support prediction of antenna performance. However, current machine learning applications in antenna design are still limited. Many studies focus on predicting only one or two parameters, such as gain or directivity, while compact MIMO antenna design requires simultaneous optimization of several parameters. Therefore, a significant research gap exists in developing generalized and multi-objective artificial intelligence models for compact mmWave MIMO antenna optimization.

Finally, although advanced materials and structures such as metasurfaces, metamaterials, flexible substrates, and additive manufacturing techniques have been increasingly investigated, their practical use in compact mmWave MIMO antennas remains limited. These technologies can improve gain, miniaturization, and beam control, but they can also introduce losses, fabrication difficulties, cost limitations, and sensitivity to manufacturing errors. Consequently, future research should focus on balancing electromagnetic performance with practical manufacturability, cost-effectiveness, and reliability.

3. Fundamentals of mmWave MIMO Systems

This section corresponds to the "Fundamentals of mmWave MIMO Systems" and "Antenna structures" branches of the taxonomy in Figure 2. It introduces the main principles, applications, advantages, and limitations of mmWave MIMO systems, before discussing the antenna structures commonly used in compact multi-port mmWave MIMO designs.

Millimeter-wave (mmWave) MIMO systems represent a fundamental technology for modern wireless communications, offering high data rates, increased spectral efficiency, and enhanced network performance. Understanding these systems requires examining several interconnected aspects, including the principles of mmWave communications, the theoretical foundations of MIMO technology, and the role of antenna structures in system design; while mmWave frequencies enable large bandwidths and support emerging applications such as 5G/6G networks and intelligent transportation systems, they also introduce challenges related to propagation losses, hardware complexity, and antenna integration. Figure 3 illustrates a taxonomy of the main concepts related to mmWave MIMO systems, emphasizing the principles of mmWave communications, MIMO system theory, antenna structures, as well as their applications, advantages, limitations, and key design challenges.

3.1. Principles of Millimeter-Wave (mmWave) Communications

Millimeter waves, or mmWave, are defined as the portion of the electromagnetic spectrum ranging from 30 to 300 GHz, corresponding to wavelengths between 1 and 10 mm [12,34,35]. Despite a strict physical definition limited to this frequency range, in the practical world, industry and standardization bodies such as 3GPP also consider bands above 24 GHz (e.g., 24.25–27.5 GHz for the n258 band and 26.5–29.5 GHz for the n257 band) as belonging to the 5G mmWave frequency range (FR2) [12]. This technology takes advantage of the underutilized millimeter wave spectrum to achieve significantly higher data rates and bandwidths than previous generations [34].

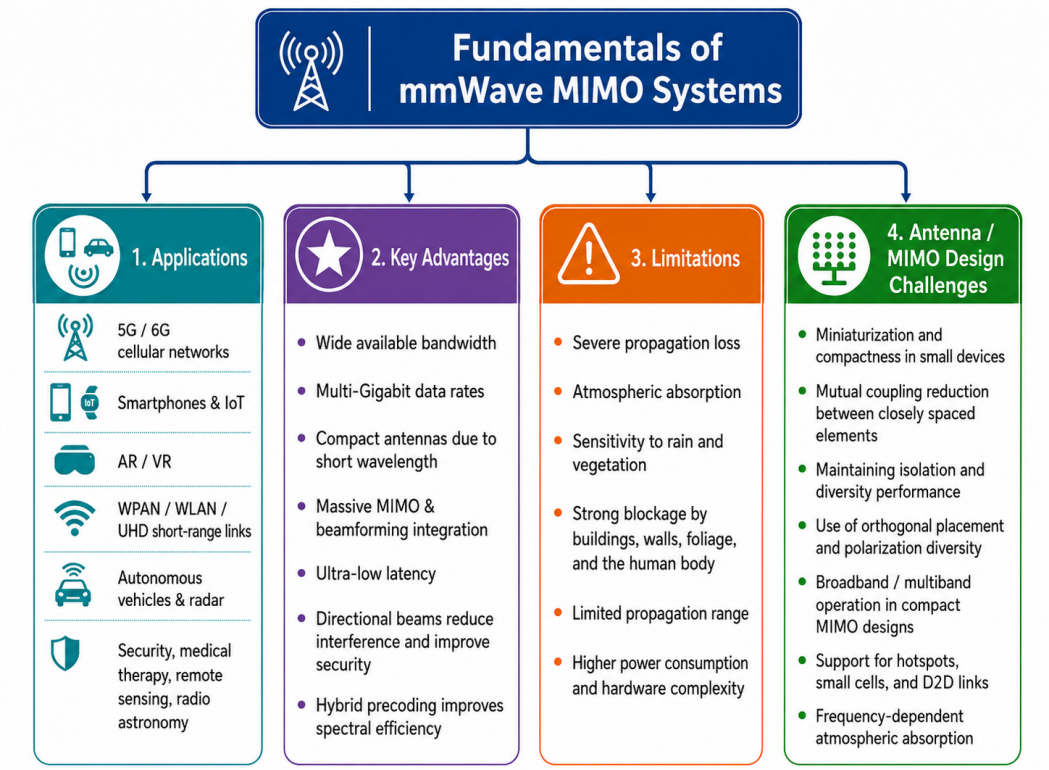


Figure 3. Taxonomy of the main concepts related to mmWave MIMO systems, including applications, key advantages, limitations, and antenna/MIMO design challenges.

Beyond telecommunications, these frequencies are widely utilized in airport security scanners, short-range radar systems, wireless personal area networks (WPAN), high-bandwidth WLANs, ultra-high-definition short-range video transmission, autonomous vehicle communications, speed-detection radar and 5G cellular networks (24–39 GHz). They are also applied in medical therapy (EHF/MMW therapy 40–70 GHz), fire-control radar (95 GHz), remote sensing, and radio astronomy. These examples highlight the versatility of mmWave technology across communications, healthcare, research, military, and security domains [35]. Furthermore, mmWave integration is increasingly adopted in smartphones and IoT devices to enable compact, high-speed connectivity [36]. It also supports immersive applications such as augmented and virtual reality (AR/VR), where ultra-high data rates and low latency are essential for real-time user experiences [37,38].

Millimeter-wave communications offer several key advantages that make them a cornerstone of next-generation wireless systems. The most significant benefit is the wide available bandwidth in the 30–300 GHz spectrum, enabling multi-gigabit data rates and supporting bandwidth-intensive applications [39–41]. The short wavelength also allows compact antenna designs, facilitating the integration of massive MIMO arrays and beamforming in small devices to greatly enhance network capacity [5,42]. In addition, mmWave systems provide ultra-low latency, which is vital for autonomous driving, industrial automation, and immersive virtual reality [43,44]. Narrow directional beams further reduce interference and improve link security, while hybrid precoding enhances spectral efficiency [45,46].

Despite these advantages, mmWave systems suffer from several inherent physical limitations. The most critical drawback is the severe propagation loss compared to sub-6 GHz frequencies, further aggravated by atmospheric absorption, rain, and vegetation [40,41]. In addition, mmWave signals are highly sensitive to obstacles such as buildings, walls, foliage, and even the human body, resulting in strong shadowing, frequent link blockages, and a

limited propagation range [5,42]. Additionally, system implementation is constrained by higher power consumption, hardware complexity, and the need for advanced architectures and protocols to ensure robust connectivity [45,46].

Integrating mmWave antennas into small wireless devices presents major challenges in miniaturization and compactness, as large antenna dimensions inherently limit their suitability for 5G systems. Reducing mutual coupling between closely spaced antenna elements is another critical challenge, since high coupling degrades isolation and diversity performance. Advanced techniques, such as orthogonal placement and polarization diversity, are often required to ensure efficient MIMO operation [7]. Broadband/multiband operation millimeter-wave technology must also support wideband cellular systems, including hotspots, small cells, and device-to-device links. This requirement highlights the difficulty of achieving broadband operation in compact MIMO antenna designs [47]. One of the major challenges in millimeter-wave systems arises from the severe propagation losses caused by atmospheric absorption, which strongly depend on the operating frequency.

3.2. MIMO System Theory

MIMO employs multiple transmitting and receiving antennas to use spatial degrees of freedom, resulting in enhanced throughput, increased connection reliability, and targeted transmission. Core mechanisms are spatial multiplexing for parallel data streams, diversity for fading mitigation, and beamforming for spatially focused signals.

3.2.1. Spatial Diversity

Spatial diversity is achieved by using multiple antennas to obtain versions of the signal affected by independent fading. This reduces the probability that deep fades occur simultaneously across all propagation paths, thus improving link robustness. This mechanism provides a diversity gain, resulting in a faster decrease in the error rate as a function of the signal-to-noise ratio (SNR), as well as a reduction in outage probability. From a theoretical perspective, the outage probability decreases with increasing diversity order and can be asymptotically approximated by a power-law function of the SNR, where the diversity order corresponds to the number of independent propagation paths. In practice, the effectiveness of spatial diversity strongly depends on the level of decorrelation between the propagation channels associated with the different antennas. However, in millimeter-wave (mmWave) systems, propagation environments are typically characterized by a limited number of paths (sparse scattering) and high directivity [48,49].

The outage probability decreases with increasing diversity order and can be asymptotically expressed as:

$$P_{\text{out}} \propto \left(\frac{1}{\text{SNR}}\right)^{d} \tag{1}$$

where d denotes the diversity order, corresponding to the number of independently faded signal paths [50].

3.2.2. Spatial Multiplexing

Spatial multiplexing, on the other hand, transmits several independent data streams concurrently over various antennas in an effort to maximize spectral efficiency. It is a potent method for high-throughput communication since the achievable capacity roughly scales with the smallest number of transmit and receive antennas. The trade-off between throughput and complexity has been managed by opportunistic and random beamforming schemes, which demonstrate that adaptive beam or user selection permits controlled multiplexing gain exploitation while lowering feedback requirements [51].

3.2.3. Beamforming

Beamforming efficiently divides a MIMO channel into parallel subchannels by using channel state information to logically combine signals in a desired spatial direction. This is essential in millimeter-wave systems where directional transmission is required to overcome high propagation losses because it suppresses interference in addition to improving the received signal power and coverage. Advanced schemes like bit-interleaved coded multiple beamforming (BICMB) demonstrate how beamforming can balance reliability and throughput by simultaneously achieving full spatial multiplexing and full diversity [52].

3.2.4. Key Performance Metrics for MIMO Systems

The diversity performance of a MIMO antenna system is commonly evaluated using several key metrics, including ECC, DG, TARC, and CCL. These parameters collectively determine how efficiently the antenna exploits spatial diversity and mitigates multipath fading effects [53]. Among these, the ECC quantifies the correlation between adjacent radiating elements. In antennas with high radiation efficiency, this parameter can be derived from the radiation characteristics. Very low levels (typically <0.02 within the operating band) indicate strong port isolation and effective MIMO performance [18]. as shown in Equation (2):

$$\rho_e = \frac{\left| \int_0^{2\pi} \int_0^{\pi} F_1(\theta,\phi) \cdot F_2(\theta,\phi)\, d\Omega \right|^2}{\left(\int_0^{2\pi} \int_0^{\pi} |F_1(\theta,\phi)|^2\, d\Omega \right) \left(\int_0^{2\pi} \int_0^{\pi} |F_2(\theta,\phi)|^2\, d\Omega \right)} \tag{2}$$

However, when the antenna exhibits high radiation efficiency, ECC can be approximated using S-parameters, which is more practical for compact MIMO antenna designs, as given in Equation (3):

$$ECC = \frac{\left| S_{ii}^* S_{ij} + S_{ji}^* S_{jj} \right|^2}{\left(1 - |S_{ii}|^2 - |S_{ij}|^2\right)\left(1 - |S_{jj}|^2 - |S_{ji}|^2\right)} \tag{3}$$

Low ECC values (e.g., <0.05) indicate weak correlation between antenna elements, ensuring good diversity performance. In many recent compact MIMO antenna designs, ECC values are maintained below 0.02 across the operating band, demonstrating excellent isolation and MIMO capability [54].

The second key metric, the DG, it can be directly calculated from ECC according to the following relation:

$$DG = 10\sqrt{1 - |ECC|^2} \tag{4}$$

A higher DG value indicates better diversity characteristics. Ideally, in many reported MIMO antenna designs, DG values greater than 9.9 dB are achieved across the operating band, confirming the effectiveness of the system [18].

The TARC is another important diversity metric. It provides an accurate characterization of the effective operating bandwidth of the entire MIMO antenna system. It also accounts for the impact of phase variations of the input signals on the antenna's impedance bandwidth [18]. Mathematically, for an N-port antenna system, TARC can be expressed in terms of incident (a_i) and reflected (b_i) waves as follows [55]:

$$\text{TARC} = \sqrt{\frac{\sum_{i=1}^{N} |b_i|^2}{\sum_{i=1}^{N} |a_i|^2}} \tag{5}$$

Using the scattering parameters (S_{ij}), and assuming random phase excitations across the ports, the TARC for a two-port MIMO antenna can be further expressed as [55]:

$$\text{TARC} = \sqrt{\frac{|S_{11} + S_{12}e^{j\theta}|^2 + |S_{22} + S_{21}e^{j\theta}|^2}{2}} \tag{6}$$

For a well-performing MIMO antenna, TARC values are typically less than 0 dB across the operating band. In reported designs, measured TARC values below −20 dB are often achieved, demonstrating excellent impedance matching and confirming the suitability of the antenna for MIMO applications [18].

The fourth key performance metric is CCL, it's used to evaluate the efficiency of MIMO antenna systems. It quantifies the reduction in the theoretical maximum channel capacity due to factors such as antenna inefficiencies, mutual coupling between elements, and spatial correlation in the propagation environment [9]. Mathematically, the CCL can be determined from the scattering parameters using the following expression [56]:

$$\text{CCL} = -\log_2 \det(\mathbf{A}) \tag{7}$$

where A is defined as the correlation matrix shown below:

$$\mathbf{A} = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix} \tag{8}$$

where:

$$\sigma_{ii} = 1 - \left(|S_{ii}|^2 - |S_{ji}|^2\right) \tag{9}$$

$$\sigma_{ij} = -\left(S_{ii}^* S_{ij} + S_{ji} S_{jj}^*\right) \tag{10}$$

Here, σ_{ii} and σ_{ij} represent the correlation coefficients between the antenna elements in the MIMO system.

Maintaining a low CCL value (typically below 0.4 bits/s/Hz) indicates efficient utilization of spatial diversity and minimal degradation in MIMO channel capacity [19].

Therefore, minimizing ECC, TARC, and CCL simultaneously remains a critical objective in the design of compact multi-port mmWave MIMO antennas.

3.3. Antenna Structures in MIMO Systems

3.3.1. Antenna Types Commonly Used

A microstrip patch antenna is a planar antenna consisting of a metallic radiating patch printed on a dielectric substrate backed by a ground plane. Due to its low-profile structure, light weight, low fabrication cost, and compatibility with printed circuit board (PCB) technology, it is well suited for compact integration in MIMO systems, including millimeter-wave and 5G applications. Such antennas are widely used in communication systems where miniaturization and multi-element integration are critical requirements [57].

However, despite their advantages, microstrip patch antennas suffer from certain inherent limitations, including a relatively narrow impedance bandwidth and performance degradation at higher frequencies due to dielectric losses and the excitation of higher-order modes, which poses a significant challenge for wideband millimeter-wave applications [58].

Recent studies have explored various microstrip patch MIMO antenna designs for millimeter-wave 5G applications, aiming to achieve compact size while enhancing MIMO performance. Cao et al. [59] demonstrated a multi-port MIMO microstrip patch antenna with metasurface-based structures, achieving wideband operation and improved diversity performance, highlighting the potential of compact 5G cellular systems. Tiwari [60] pro-

posed a two-element microstrip-fed slot-loaded MIMO antenna with Y-shaped elements and slot loading to improve impedance bandwidth and isolation, suitable for 5G mmWave communications. Jeong [61] introduced a microstrip patch antenna employing a vertically coupled split-ring metaplate to enhance gain and bandwidth, providing a compact solution for mmWave systems. Farghaly et al. [62] designed a 28 GHz microstrip patch MIMO antenna with multiple slots to optimize impedance matching and radiation characteristics, demonstrating robust performance for IoT and beyond 5G networks. The geometry of the proposed antenna is illustrated in Figure 4.

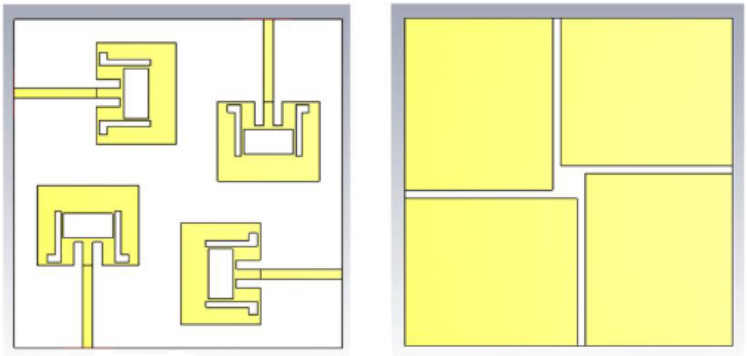


Figure 4. Configuration of the proposed 4 × 4 MIMO antenna [62].

Al-Hiti et al. [63] developed broadband and multiband microstrip patch antennas on low-cost substrates, achieving compact designs compatible with high-capacity mmWave systems and phased array integration. Maharjan and Choi [64] proposed a four-element microstrip patch array with a combined corporate-series feeding network, improving bandwidth and gain for 5G communications. Abhishek and Suraj [65] presented an aperture-coupled feed microstrip patch antenna achieving miniaturization and enhanced bandwidth at 39 GHz, suitable for compact mmWave devices. Tadesse et al. [66] designed a planar four-port MIMO antenna with split-ring resonators to introduce additional resonance bands, achieving excellent isolation, low ECC, and high diversity gain for 28/38 GHz 5G applications. Finally, Sethi et al. [67] proposed a dual-band four-element MIMO patch antenna with pattern diversity and slot modifications, delivering robust MIMO performance with low ECC and high diversity gain for mmWave 5G networks.

Dielectric Resonator Antennas (DRAs) constitute a class of microwave antennas that utilize a block of high-permittivity dielectric material, typically cylindrical, rectangular, or hemispherical in shape, placed on a conductive ground plane. DRAs operate by exciting resonant modes within the dielectric resonator, enabling efficient radiation through the dielectric–air interface. Due to their operating principle, DRAs inherently offer compact size, low loss, and comparatively wide impedance bandwidth. The physical dimensions of a DRA are proportional to the free-space wavelength divided by the square root of the dielectric constant, which enables significant miniaturization when dielectric materials with permittivity values in the range of 10–100 are employed [68]. Unlike conventional metallic antennas, DRAs do not rely on metallic radiating elements, thereby eliminating conductor losses and achieving high radiation efficiency, which is particularly advantageous at millimeter-wave frequencies. Furthermore, DRAs can be efficiently integrated with planar feeding structures such as microstrip or stripline through simple coupling mechanisms, while their bandwidth characteristics can be flexibly controlled by adjusting the dielectric constant and resonator geometry [69].

Recent studies have achieved the strong potential of DRAs for mmWave applications due to their high radiation efficiency, wide bandwidth, and low-loss characteristics. Multiple techniques, including deep learning optimization, metallic vias, 3D printing, complementary split-ring resonators (CSRRs), and hybrid resonator configurations, have been explored to improve bandwidth, gain, isolation, and compactness in both single-element and MIMO DRA designs [70–73]. Several reported designs achieve good MIMO performance metrics and enhanced radiation characteristics while supporting wideband or dual-band operation in the 28 GHz and 39 GHz 5G bands [72–74]. However, many of these improvements rely on complex geometries, hybrid structures, or advanced fabrication techniques such as 3D printing and multi-resonator integration, which increase the overall design and manufacturing complexity [71,73,74]. In addition, some designs remain limited to single-element or narrowband configurations, restricting their suitability for compact multi-port MIMO systems [70,75,76]. Figure 5 shows representative dual-band DRA configurations employing metallic vias and 3D-printed encapsulated structures for mmWave 5G applications [71]. Overall, the literature highlights DRAs as promising candidates for future 5G mmWave MIMO systems, although achieving compact multi-port integration with simplified structures and wideband performance continues to be a significant challenge.

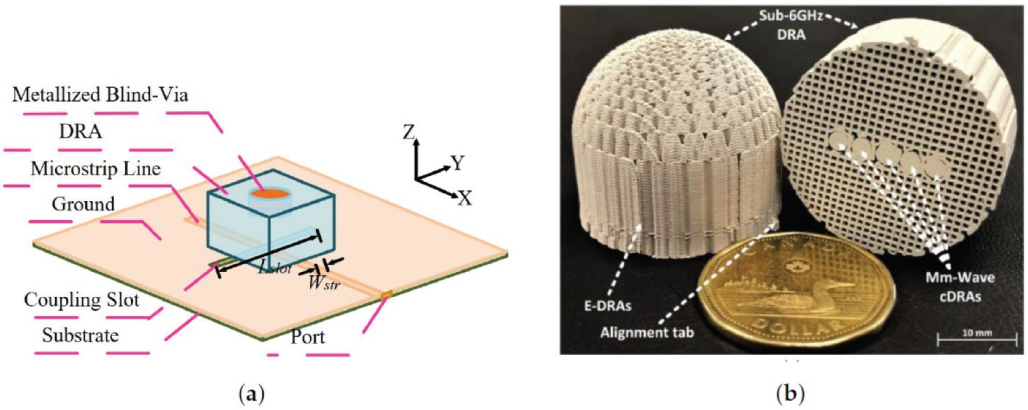


Figure 5. Configuration of the dual-band DRAs covering mmWave bands: (**a**) PCB-based multi-mode DRA with metallic via; (**b**) 3D-printed encapsulated DRA [71].

The slot antenna is an adaptable antenna that consists of a small slit in a ground plane. It has been utilized in all facets of radar and wireless applications and may be modified to work with waveguide, coplanar waveguide (CPW), coaxial, slot line, or microstrip feeding schemes. The planar slot antenna exhibits bidirectional radiation, which restricts its use in contrast to the waveguide slot antenna, which is energized from one side. At the expense of extra bulk and complexity, the slot can also be cavity- or reflector-backed to remove backside radiation. The basic building block of a slot antenna is a microstrip transmission line that electromagnetically excites a narrow, straight slot. Although this straightforward configuration is easy to integrate and has a planar design, resonance requirements usually limit its physical length, which is frequently close to half the wavelength [77]. Several changes to the slot geometry, including meandering, folding, or adding more perturbations, have been made to satisfy the requirements of miniaturization. These methods allow for smaller antennas with respectable radiation and impedance matching. Shafai et al. have published in-depth research on the design concepts and performance trade-offs of such antennas [78,79].

Recent studies on slot-based MIMO antennas for mmWave applications collectively demonstrate that slot engineering, combined with techniques such as defected ground structures (DGSs), polarization diversity, fractal geometries, and substrate-integrated waveguide

(SIW) configurations, can significantly enhance bandwidth, isolation, and diversity performance. Most reported designs ensure good MIMO performance metrics, confirming the suitability of slot-based structures for high-capacity mmWave communications [80–83].

Figures 6 and 7 presents representative examples of slot-based mmWave MIMO antenna structures. These examples show how slot loading and ground-plane modifications can be used to obtain compact radiating elements.

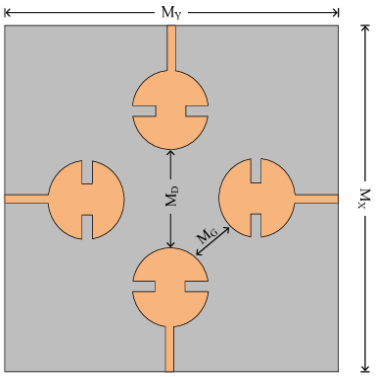


Figure 6. Slot-based mmWave MIMO antenna using a defected ground structure [80].

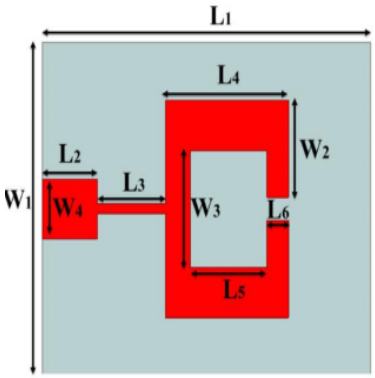


Figure 7. Representative slot-loaded mmWave MIMO antenna configuration [84].

Several works further improved port isolation and impedance matching through advanced slot geometries and optimization approaches, including inverted I-shaped slots, circular and star-shaped slots, and neural network-assisted optimization techniques [84,85]. Nevertheless, these improvements are often accompanied by important limitations, such as increased antenna size, higher geometric complexity, complex ground-plane structures, or hybrid architectures that may hinder compact integration [82–85]. Overall, the study shows that slot-based MIMO antennas are a good way to make things small and work well at mmWave frequencies. However, finding the best balance between making things smaller, making them simpler, and improving MIMO performance is still a big problem that needs to be solved.

Compact antennas based on metamaterials are a broad class of resonant structures that use artificial inclusions and engineered dispersion to accomplish features that are not possible with traditional designs. Composite right/left-handed (CRLH) transmission line antennas are one of the most researched configurations; in these antennas, electrically small resonators can radiate effectively through the synthesis of negative-order modes and zeroth-order resonators. Specifically, zeroth-order resonance makes it possible to separate the physical dimensions of the antenna from the resonant wavelength. Recent studies, such as [86], have further demonstrated how tunable ZOR designs can enhance compactness and adaptability. Another important strategy is based on metamaterial

loadings, which include high-permeability shells, the radiator, and epsilon-negative (ENG), mu-negative (MNG), or double-negative (DNG) materials. For instance, ref. [87] illustrates how carefully engineered DNG structures can improve antenna performance while enabling miniaturization. In this category, Ziolkowski's near-field resonant antenna (NRA) concept has shown that subwavelength inclusions can couple to free-space modes in a strong way, allowing for efficient radiation from structures much smaller than $\lambda/2$.

Metaresonator antennas, which are usually based on split-ring resonators (SRRs) and complementary SRRs (CSRRs), offer a further extension. They offer improved quality factor (Q), sharp frequency selectivity, and flexible miniaturization techniques. Because the resonant inclusions in these designs can be tuned or cascaded to create multiple passbands in a small footprint, they are especially appealing for applications that need dual-band or multiband operation [88]. Lastly, metasurface-loaded antennas, like those that use reactive impedance surfaces (RIS) or electromagnetic bandgap (EBG) structures, provide extra advantages like surface wave suppression, increased bandwidth, improved front-to-back radiation ratio, and even low-profile integration with planar circuits. Recent work such as [89] further shows how CRLH/ZOR designs coupled with metasurface or network feeding (like a Butler matrix) can yield compact, efficient antennas with good gain and control. All of these advancements show that metamaterials have not only solved the basic size–efficiency constraints of traditional antenna theory, but have also given designers a platform to create small, multipurpose antennas that are appropriate for new wireless, biomedical, and sensing applications.

Several recent studies have investigated the use of metamaterials to enhance the performance of mmWave MIMO antennas for 5G applications. In [90], a compact dual-port MIMO antenna is proposed, where a 4×1 metamaterial array is inserted between two mirror-inverted radiating elements to suppress surface waves, thereby improving isolation, gain, and antenna miniaturization; however, the design is limited to a two-port configuration. In [91], a tri-band compact MIMO antenna targeting 5G IoT applications employs an epsilon-negative (ENG) metamaterial to control electromagnetic field distribution, leading to enhanced isolation and gain across multiple frequency bands, at the cost of increased design complexity due to the multi-band operation. In [92], a quad-port multi-band mmWave MIMO antenna integrates metamaterial decoupling cells as shown in Figure 8 to significantly reduce mutual coupling, resulting in high isolation and excellent diversity performance, although the overall antenna size remains relatively large. A different approach is presented in [93], where a metasurface with negative effective permittivity and permeability is used as a superstrate above a dielectric resonator antenna to enhance gain and directivity; nevertheless, the suspended metasurface may introduce fabrication and mechanical challenges. Finally, ref. [59] utilizes a square-ring resonator (SQRR) metasurface printed on a microstrip patch antenna, acting as a secondary radiator to achieve wideband mmWave operation and improved gain, while full experimental validation is still required.

3.3.2. Single-Element Performance Metrics

A single antenna element's performance is often assessed using a set of critical parameters that establish its appropriateness for a particular application. Effective power transfer between the feed and the radiating element is ensured by the impedance bandwidth, which measures the frequency range throughout which the antenna maintains adequate impedance matching (usually $|S_{11}| \leq -10$ dB). This is closely connected to radiation efficiency, which is the percentage of input power that is actually radiated after accounting for dielectric and ohmic losses; recent studies have investigated advanced approaches for efficiency measurement in reverberation chambers [94]. As a crucial measure of connection performance, the gain combines efficiency and radiation directivity, signifying the

antenna's capacity to concentrate radiated power in a particular direction in contrast to an isotropic source. Structural loading techniques such as split-ring resonators (SRR) and fractal geometries have been successfully applied to achieve multiband operation and improve radiation characteristics in compact antennas [95]. The radiation pattern describes the angular distribution of radiated energy. It can be either omnidirectional, radiating uniformly in the azimuth plane for applications like broadcasting or mobile connectivity, or directional, concentrating energy in a preferred direction for long-range communication. Recent work on quasi-isotropic radiation using innovative current distribution strategies illustrates ongoing progress in pattern synthesis [96]. When combined, these metrics offer a thorough framework for evaluating and contrasting single-element antennas across various design approaches and functional needs.

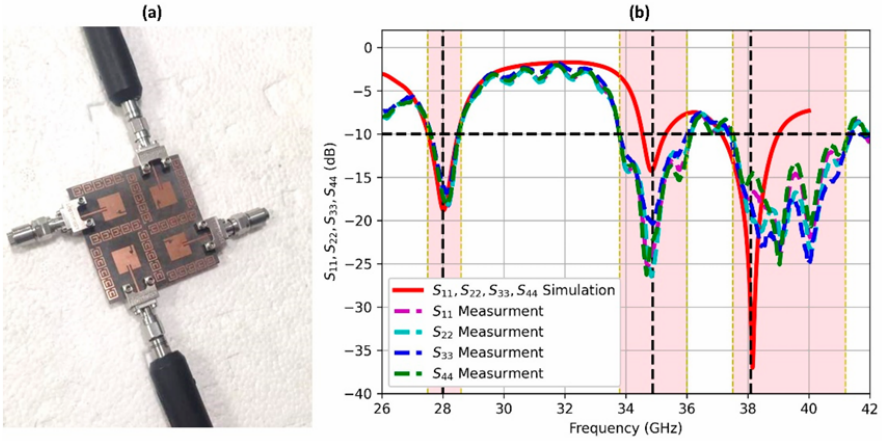


Figure 8. (**a**) Four-port MIMO prototype; (**b**) measured and simulated reflection coefficients for each MIMO [92]. The black dashed horizontal line represents the −10 dB criterion, while the vertical dashed lines and the pink shaded areas indicate the relevant operating frequency bands, in accordance with the original source.

4. Design Techniques for Compact Multi-Port mmWave MIMO Antennas

Compact mmWave MIMO antennas face several key challenges, including miniaturization, high mutual coupling between closely spaced elements, limited radiation efficiency, and the demand for wideband operation [7,47]. To address these issues, a wide range of design techniques has been reported in the literature [2], including mutual coupling reduction methods, miniaturization strategies, bandwidth and gain-enhancement techniques, advanced feeding networks, as well as the integration of artificial materials and device-level electromagnetic structures. This section corresponds to the "Design techniques" branch of the taxonomy in Figure 2.

To make the comparison more rigorous, the reviewed antenna structures and decoupling techniques are analyzed using a common quantitative framework. The comparison considers operating frequency, number of ports, bandwidth, gain, isolation, ECC, radiation efficiency, antenna size when available, and implementation complexity. These criteria allow the different techniques to be evaluated not only according to their reported performance values, but also according to their practical trade-offs in terms of compactness, fabrication sensitivity, scalability, and integration feasibility. Therefore, the discussion following each comparative table highlights the strongest performance indicators, the main limitations, and the most relevant application scenarios for each technique.

4.1. Mutual Coupling Reduction Approaches

As indicated in the mutual coupling reduction branch of Figure 2, this subsection focuses on the main decoupling strategies used in compact mmWave MIMO antennas,

including DGS, parasitic and decoupling elements, polarization/spatial diversity, and special material-based approaches.

Mutual coupling in MIMO antenna systems refers to the electromagnetic interaction between closely spaced antenna elements, primarily caused by near-field coupling, surface–current propagation, and surface–wave excitation along the substrate. This interaction alters the input impedance, reflection coefficients, and radiation patterns, leading to performance degradation, including reduced SINR, distorted channel estimation, and decreased diversity and capacity in MIMO systems [97]. To mitigate these adverse effects, a variety of decoupling techniques have been proposed in the literature, all of which have demonstrated effectiveness in improving isolation and reducing correlation.

Physically, mutual coupling reduction is achieved by controlling how electromagnetic energy is transferred from one antenna element to another. In compact mmWave MIMO antennas, this unwanted transfer generally occurs through direct near-field interaction, surface currents flowing on the common ground plane, and substrate-supported surface waves [97,98]. Therefore, effective decoupling structures generally act by interrupting the dominant current path, suppressing surface–wave propagation, creating an opposite-phase coupling path, or forcing the antenna elements to radiate through weakly correlated field components [99,100]. This physical interpretation is important because two techniques may provide similar isolation values while affecting bandwidth, gain, radiation efficiency, compactness, and fabrication tolerance in different ways [98].

In the following subsections, each technique and its variants are discussed in detail.

More specifically, DGS and slot-based ground modifications disturb the current flow on the ground plane and increase the effective path between adjacent antenna elements, which reduces near-field coupling and surface–current interactions [98–100]. EBG, FSS, and metamaterial-based structures operate as stop-band or resonant filtering surfaces that inhibit surface–wave propagation within the operating band [98,101]. Neutralization lines and parasitic decoupling elements create additional coupling paths with opposite phase, which can cancel part of the undesired coupled current between antenna ports [98,102]. Finally, polarization and spatial diversity reduce mutual coupling by exciting orthogonal field components or increasing the effective electrical separation between radiating elements, thereby decreasing the correlation between MIMO channels [98,103].

4.1.1. Defected Ground Plane (DGS)

In multi-port MIMO antennas, DGS reduces mutual coupling by suppressing surface–wave propagation and interrupting undesired current paths between adjacent radiating elements. This leads to enhanced port isolation, reduced envelope correlation coefficient (ECC), and improved diversity performance [8,104,105]. Initially introduced in simple dumbbell-shaped forms, DGS has since evolved into a wide range of geometries, including rectangular, circular, U-, V-, H-shaped, spiral, fractal, concentric-ring, and split-ring resonator (SRR) configurations, allowing designers to target specific objectives such as antenna miniaturization, bandwidth enhancement, or radiation control [99,100].

At the physical level, a DGS acts as a controlled discontinuity in the ground plane. When surface currents propagate between adjacent antenna elements, the etched defect disturbs their natural path and forces them to follow a longer or more confined route. This modifies the equivalent inductance and capacitance of the ground plane and can create a resonant stop-band effect around the operating frequency, which reduces surface–wave propagation and weakens the coupling path between ports [98–100]. The decoupling behavior is strongly geometry-dependent: simple rectangular or slot-shaped DGS mainly interrupt and lengthen the current path, circular or ring-shaped defects introduce resonant

current loops, while zigzag, fractal, or star-shaped geometries create multiple discontinuities that can enhance or broaden the suppression of coupled currents [99,100].

From a performance standpoint, DGS-based designs have demonstrated significant isolation enhancement in compact mmWave MIMO antennas, with reported isolation levels often exceeding 30 dB and, in optimized cases, reaching values above 40 dB [8,104]. Such isolation enhancement contributes to improved signal quality, higher diversity gain, and reduced inter-port interference. In addition, DGS can facilitate impedance matching and bandwidth enhancement, while maintaining a compact footprint. DGS offers advantages in terms of structural simplicity, compactness, fabrication ease, and cost-effectiveness [99,100]. However, the impact of DGS on radiation efficiency and gain is highly dependent on the defect geometry and operating frequency, particularly at millimeter-wave bands.

Despite these advantages, DGS-based approaches present certain limitations. Their performance is highly sensitive to the geometry, size, and placement of the defects, making design optimization challenging and potentially leading to additional losses or frequency instability if not carefully engineered [99]. Moreover, at millimeter-wave frequencies, fabrication tolerances become increasingly critical, as small deviations in defect dimensions can significantly affect antenna performance [100]. In some configurations, extensive ground-plane modification may also compromise mechanical robustness or limit integration in highly compact or conformal platforms.

Recent studies have demonstrated the effectiveness of defected ground structures (DGSs) in enhancing the performance of mmWave MIMO antennas. Therefore, the use of DGS in both single-band and dual-band mmWave MIMO antennas can be considered a clear trend in the reviewed literature. The reported results, as shown in Table 3, indicate that DGS-based designs can achieve high isolation levels, sometimes exceeding 30–40 dB and reaching even higher values in optimized cases, with low ECC and favorable MIMO diversity metrics [56,80,106–108].

Another important note is that DGS is frequently combined with other isolation mechanisms to further improve MIMO performance. Hybrid techniques employing spatial diversity, polarization diversity, metallic sidewall strips, or dielectric resonator antenna configurations have been suggested to enhance isolation, bandwidth, and radiation stability. These combinations can lead to very high isolation and wideband operation, but also increase the structural complexity of the antenna. This makes the design more sensitive to fabrication tolerances, especially at millimeter-wave frequencies where small dimensional variations can strongly affect the measured performance [56,73,82]. Representative examples of these studies are summarized in Table 3.

Figure 9 presents representative DGS geometries from selected studies reported in Table 3. These examples show the evolution from slot-based and rectangular DGS configurations toward more complex geometries designed for high-isolation mmWave MIMO antennas.

Table 3 indicates that DGS-based techniques are effective for reducing mutual coupling in compact mmWave MIMO antennas. Most of the reviewed designs achieve isolation levels higher than 30 dB, while maintaining low ECC values. This confirms that DGS can improve port isolation and diversity performance in compact MIMO configurations.

From a quantitative point of view, ref. [56] reports the highest gain, reaching 12 dBi at 28 GHz, with isolation higher than 40 dB and ECC below 0.0003. The strongest isolation performance is reported in [107], with isolation levels higher than 39 dB and 60 dB at the 28/38 GHz bands, together with an ECC below 0.00005. These results indicate that optimized DGS placement between radiating elements can provide very strong coupling reduction in dual-band MIMO designs.

In terms of frequency coverage, ref. [82] covers the widest operating range, from 26.5 to 40 GHz, using a partial ground with circular and star-shaped slots. However, this design is reported as simulated, whereas several other studies in the table include both simulated and measured validation. Therefore, experimental validation remains important when evaluating the practical robustness of wideband DGS-based designs.

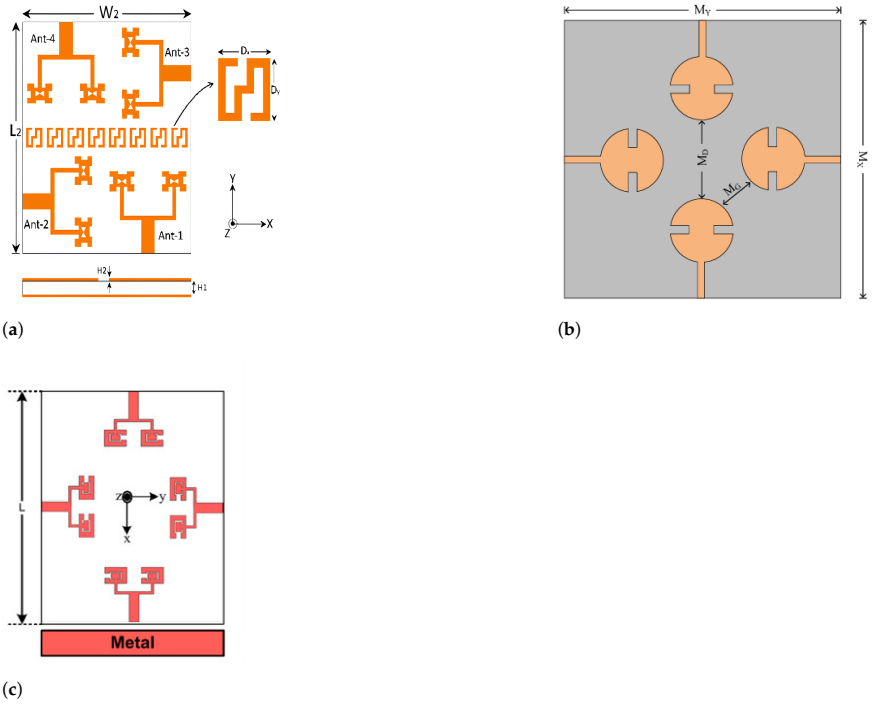


Figure 9. Representative DGS geometries employed in selected mmWave MIMO antenna designs. (**a**) DGS configuration reported in [56]. (**b**) Iterative rectangular DGS reported in [80]. (**c**) Slot-based DGS configuration reported in [106].

Table 3. Representative mmWave MIMO antennas employing defected ground structures (DGSs).

Ref.	Data Type	Freq. (GHz)	Ports	Substrate	Gain (dBi)	Bandwidth (GHz)	Isolation (dB)	ECC	DGS Geometry
[56]	Measured/ Simulated	28	4	Rogers RT 5880	12	1	>40	<0.0003	Vertical and horizontal ground slots
[80]	Measured/ Simulated	28	4	Rogers RT/5880	7.1	4	>−30	0.0005	Iterative rectangular DGS
[82]	Simulated	26.5–40	4	Rogers RT/Duroid 5880	7.17	5.37	>−50	<0.02	Partial ground with circular and star-shaped slots
[106]	Measured/ Simulated	25.5–29.6	4	Rogers substrate	8.3	4.1	>30	<0.01	Rectangular, circular, and zigzag slots
[107]	Measured/ Simulated	28/38	4	Rogers TMM4	9	Dual-band	<−39/−60	<0.00005	Ground DGS between perpendicular patches
[108]	Measured/ Simulated	28/38	2	Rogers substrate	6.2/7	Dual-band	−40/−42	0.0015	Slot-based DGS in partial ground

The comparison also shows that DGS geometries vary from relatively simple rectangular or slot-based configurations to more complex circular, zigzag, star-shaped, and hybrid ground modifications. Simple DGS structures, such as those reported in [80,106,108], provide a practical balance between isolation improvement and structural simplicity. More complex DGS configurations, such as those in [82,107], can achieve stronger isolation or wider frequency coverage, but they require more careful geometrical optimization.

Overall, the quantitative comparison shows that DGS-based decoupling is a strong candidate for compact mmWave MIMO antennas. However, the choice of DGS geometry should depend on the targeted trade-off between isolation, gain, bandwidth, ECC, number of ports, and validation level. Therefore, a DGS should not be evaluated only by its maximum isolation value, but by its overall balance between electromagnetic performance and practical implementation.

4.1.2. Decoupling Elements and Parasitic Structures

Decoupling elements play a vital role in enhancing the performance of compact MIMO antennas. To address the issue of mutual coupling, various decoupling techniques have been proposed, including isolation networks and parasitic structures. These elements introduce additional coupling paths or disturb the surface current distribution, thereby suppressing or compensating the undesired interactions between antenna elements. As a result, the radiation characteristics, isolation level, and efficiency of MIMO antennas can be significantly improved without increasing the physical footprint [101,102].

Isolation networks constitute a widely adopted class of decoupling techniques for compact MIMO antennas, particularly when strong electromagnetic coupling arises from closely spaced radiating elements [109,110]. These networks are typically implemented using transmission-line-based structures or lumped elements, such as decoupling/matching networks (DMNs), inductors, stop-band filters, and neutralization lines, inserted between antenna elements or feeding lines. Their operating principle relies on the creation of controlled counter-phase current paths that compensate for undesired coupling, thereby significantly enhancing port isolation while preserving radiation performance [111].

Physically, isolation networks and neutralization lines reduce coupling by controlling the amplitude and phase of the current induced between adjacent antenna elements. When one port is excited, part of the electromagnetic energy is normally coupled toward the neighboring port through near fields and surface currents. A properly designed neutralization line introduces an additional coupling path whose induced current is approximately opposite in phase to the undesired coupled current. This cancellation reduces the net current reaching the adjacent port and therefore lowers the transmission coefficient between ports. However, this mechanism is frequency-dependent because the phase compensation depends on the electrical length, position, and impedance of the added line or network [102,111,112].

Among these approaches, neutralization lines represent one of the simplest and most intuitive implementations of isolation networks. By electrically connecting adjacent radiators through metallic strips or transmission lines, neutralization lines generate canceling coupling paths that effectively suppress mutual coupling. Their effectiveness has been demonstrated in various MIMO antenna configurations, including millimeter-wave systems [112]. A representative example is reported in [113], where a UWB Vivaldi MIMO antenna was designed for millimeter-wave 5G applications.The design uses both neutralization lines and CSRR metamaterial cells to lower the coupling between parts that are close to each other. However, at millimeter-wave frequencies, the performance of this approach can be sensitive to fabrication tolerances and additional losses, which may limit their scalability in highly integrated multi-port MIMO arrays.

Parasitic decoupling structures, on the other hand, act as passive electromagnetic barriers placed near or between radiators. Structures interact with antenna currents to cancel or redirect undesired coupling [114].

At the physical level, parasitic decoupling structures operate through induced currents. Since these elements are not directly fed, they are excited by the near field of the active radiating elements. Their geometry and position are selected so that the induced current

either cancels part of the coupled current, redirects the surface–current flow away from the neighboring antenna, or creates a local resonant barrier between ports. For example, X-shaped or cross-shaped strips provide diagonal current paths that oppose the dominant coupling direction, while T-shaped stubs modify the ground current distribution and can introduce a localized impedance discontinuity. This explains why parasitic structures can improve isolation without requiring a large increase in antenna footprint, although their performance remains sensitive to dimensional tuning at mmWave frequencies [114–116].

For example, the X-shaped decoupling strip is a parasitic structure introduced in compact microstrip antenna arrays to reduce mutual coupling, particularly when the inter-element spacing is smaller than $\lambda_0/2$. The X-shaped geometry provides alternative current paths that suppress undesired electromagnetic interactions, leading to improved isolation and enhanced impedance matching [115].

In the design described by [117], an X-shaped decoupling element was incorporated into a small quad-port MIMO antenna for 5G FR2 applications. Its diagonally oriented arms produced alternate current paths to reduce surface current interaction and enhance isolation performance. Other similar approaches of using parasitic strips were evaluated for use in DRA MIMO systems, including the cross-shaped metallic parasitic strip described in [118], which improved isolation and successfully changed current distribution. However, these techniques showed improved radiation isolation and efficiency; some designs still exhibit moderate gain performance and increased structural complexity.

T-shaped decoupling elements, on the other hand, are implemented as stubs in the ground plane. Beyond isolation improvement, these structures can also enhance impedance matching and radiation efficiency [116]. For example, in [119], a T-shaped ground stub was inserted between closely spaced MIMO elements. The proposed configuration achieved isolation levels exceeding 20 dB while maintaining stable radiation patterns and improved impedance matching characteristics. However, the effectiveness of the T-shaped structure strongly depends on precise dimensional tuning, which may increase the design complexity and sensitivity to fabrication at higher frequencies.

The application of parasitic and isolation network techniques in true millimeter-wave (mmWave) MIMO systems remains relatively limited in the literature, despite their demonstrated effectiveness at sub-6 GHz and UWB frequencies. At mmWave frequencies, the reduced wavelength significantly increases dimensional sensitivity, fabrication tolerances, and conductor/dielectric losses, which may degrade the performance of parasitic strips, neutralization lines, and ground stubs. Moreover, the incorporation of these structures in densely packed multi-port arrays functioning beyond 24 GHz presents further hurdles associated with coupling mechanisms primarily influenced by surface waves and near-field interactions. Consequently, systematic investigations of these decoupling strategies in mmWave MIMO antennas remain scarce, particularly for scalable multi-port configurations.

Table 4 compares representative parasitic decoupling structures and isolation networks used in compact mmWave MIMO antennas. The reported results show that all reviewed designs achieve isolation levels higher than 22 dB, while maintaining low ECC values. This suggests that parasitic strips, neutralization lines, CSRR cells, and ground stubs can effectively reduce mutual coupling between closely spaced antenna elements.

From a quantitative point of view, the highest gain among the listed examples is reported in [113], reaching 13.5 dBi. This design also provides isolation higher than 30 dB and a very low ECC of 6.4×10^{-8}, which indicates strong diversity performance. However, this result is obtained using a hybrid decoupling approach combining neutralization lines with CSRR metamaterial cells, which makes the structure more complex than single parasitic-strip or ground-stub configurations.

In terms of operating frequency coverage, ref. [113] covers the widest range, extending from 26 to 40 GHz. Among the four-port designs, ref. [117] reports a bandwidth of 6.7 GHz over the 22.5–29.2 GHz range. This design uses an X-shaped decoupling structure combined with multi-slit and multi-slot techniques, achieving isolation higher than 27 dB and an ECC of 0.004. In comparison, refs. [118,119] provide narrower bandwidths of approximately 4 GHz and 2 GHz, respectively, while maintaining isolation levels higher than 25 dB and 22 dB.

Table 4. Comparative summary of decoupling techniques using parasitic structures and isolation networks.

Ref.	Data Type	Freq. (GHz)	Ports	Substrate	Gain (dBi)	Bandwidth (GHz)	Isolation (dB)	ECC	Decoupling Technique
[117]	Measured/ Simulated	22.5–29.2	4	Rogers RT/ Duroid 5880	4.25	6.7	>27	0.004	X-shaped decoupling structure with multi-slit and multi-slot techniques
[113]	Measured/ Simulated	28–38	2	Rogers RT/ Duroid 5880	13.5	26–40	>30	6.4×10^{-8}	Neutralization lines combined with CSRR metamaterial cells
[118]	Measured/ Simulated	26–30	2	Rogers RT/ Duroid 5880	8–10	≈4	>25	<0.01	Cross-shaped metallic parasitic strips with DGS
[119]	Measured/ Simulated	27.5–29.5	2	Rogers RT/ Duroid 5880	8.4	2	>22	<0.02	T-shaped ground stub inserted between MIMO elements

The comparison shows that hybrid decoupling techniques can provide strong performance in terms of gain, isolation, and ECC, but usually at the cost of increased structural complexity. In contrast, simpler parasitic or ground-stub structures offer more direct implementation and may be more suitable for compact and fabrication-conscious MIMO designs. Therefore, the choice of decoupling technique should not be based only on the maximum isolation value. It should also consider bandwidth, gain, ECC, number of ports, structural complexity, and the validation type reported in each study.

4.1.3. Polarization and Spatial Diversity

Polarization is a key parameter in antenna design, especially for MIMO systems where polarization diversity can improve isolation and channel capacity without enlarging the antenna footprint [120]. Different polarization methods can be used based on the direction and properties of the fields that are sent out.

Linear polarization (LP) is when the electric field that is transmitted moves in a single direction, usually horizontally or vertically [121]. Orthogonal linear polarizations are frequently used in MIMO and dual-polarized antennas to enhance isolation and allow independent signal transmission between antenna ports [122].

The ±45° slant polarization, on the other hand, refers to linear polarization oriented at +45° or −45° with respect to the horizontal and vertical axes. In wireless communications, ±45° polarization enhances system capacity and link reliability through polarization diversity, widely adopted in 5G millimeter-wave systems [123].

As for circular polarization, it reduces losses due to antenna misalignment and mitigates Faraday rotation in satellite and terrestrial links [121]. In practice, most antennas generate imperfect circular polarization, leading to elliptical polarization, which is quantified using the axial ratio. Common antenna structures for circular polarization include crossed dipoles, helical antennas, quadrifilar, and spiral designs [124].

The polarization diversity is further increased by combining circular and linear polarizations. In compact MIMO designs, circular polarization complements linear polarization by reducing multipath fading, interference, and orientation sensitivity. Integrating CP/LP polarization diversity enhances port isolation, reduces signal correlation, and upgrades

robustness, making it a strongly attractive solution for compact and reliable 5G wireless devices [2,125].

From a physical point of view, polarization diversity reduces coupling because orthogonally polarized antenna elements radiate electric-field components in different orientations. As a result, the field produced by one element is weakly received by the other element when their polarizations are orthogonal or sufficiently different. This reduces the overlap between radiation patterns, lowers the envelope correlation coefficient, and improves MIMO diversity performance without requiring a large physical separation between ports [103,120,122]. Circular or slant-polarized elements can also improve link robustness by reducing orientation sensitivity and multipath-induced polarization mismatch, which is particularly useful in compact mmWave terminals [121,123,125].

Spatial diversity is one of the most widely adopted techniques to improve the reliability and performance of MIMO systems. It relies on deploying multiple radiating elements in distinct positions to reduce correlation and mitigate signal degradation caused by fading. Sufficient spacing between antenna elements minimizes mutual coupling and ensures the independent reception of signals. In compact devices where physical space is limited, the orientation and arrangement of antennas play a critical role: configurations such as face-to-face placement, diagonal arrangements, or 180° rotations of elements can optimize isolation and maintain effective diversity [126].

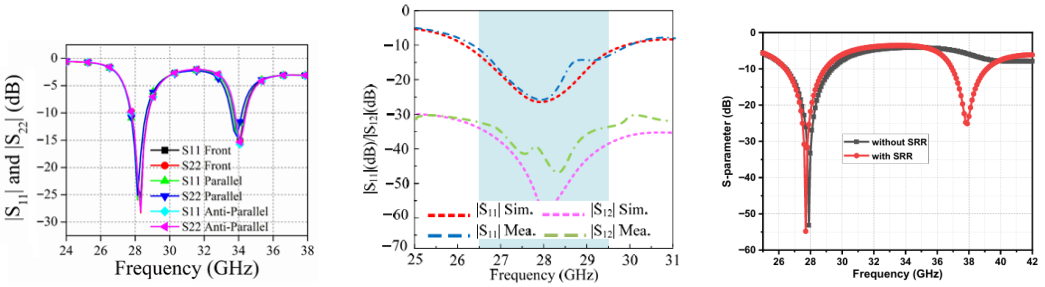
Physically, spatial diversity reduces mutual coupling by increasing the effective electrical separation between radiating elements or by orienting them so that their strongest near-field regions do not strongly overlap. Even when the physical spacing is limited, diagonal placement, opposite orientation, or 180° rotation can redirect the surface–current distribution and reduce the direct coupling path between ports. This leads to lower channel correlation, improved isolation, and reduced channel capacity loss, while preserving compact antenna dimensions [103,126,127].

This concept is particularly crucial in millimeter-wave (mmWave) systems, where stringent size constraints demand careful optimization of antenna architectures. The effectiveness of advanced MIMO techniques, such as spatial multiplexing, depends directly on spatial decorrelation: "For SM to work well, the channel must provide sufficient decorrelation between the different closely spaced antennas." In compact 5G terminals, limited physical separation requires innovative engineering approaches that combine strategic element placement (e.g., at board corners), orthogonal orientation, and the synergistic use of polarization diversity. These arrangements maximize the effective "electrical distance" between ports, reducing correlation and minimizing channel capacity loss (CCL), while preserving the minimal footprint required for next-generation mobile devices [103].

Combining spatial and polarization diversity improves link reliability and lowers channel correlation in 5G compact mmWave MIMO antennas. Farahat and Hussein [127] show that while cross-polarized elements offer polarization diversity, co-polarized arrangements take use of spatial diversity through element spacing and geometry. This dual strategy demonstrates the advantages of combining both diversity strategies in a compact design by guaranteeing low ECC, high DG, and constant efficiency across several frequency bands (28, 43, 52, and 57 GHz).

Several recent studies have demonstrated the implementation of polarization and spatial diversity in compact MIMO antennas for millimeter-wave. The efficiency of diversity strategies in lowering signal correlation and improving link reliability is confirmed by the majority of documented systems, which reach very low ECC values and diversity gains around the theoretical limit of 10 dB [66,80,84]. The increasing use of machine learning techniques, such as neural networks and predictive models, to optimize antenna parameters and speed up the design process [84,128] is another noteworthy trend. Figure 10 illustrates

representative scattering−parameter responses reported in compact 5G mmWave MIMO antennas. In [84], the different antenna orientations, namely front, parallel, and anti-parallel configurations, maintain good impedance matching around the targeted millimeter-wave bands. In [80], the comparison between simulated and measured S_{11} and S_{12} results shows good agreement around the 28 GHz band, confirming the impedance matching and low mutual coupling of the proposed MIMO antenna. Similarly, the use of SRR loading in [66] introduces an additional resonance around 38 GHz while maintaining the initial resonance near 28 GHz. This demonstrates how resonant structures such as SRR or CSRR can be used to achieve dual-band operation and enhance compact MIMO antenna performance.



(**a**) (**b**) (**c**)

Figure 10. Examples of scattering−parameter responses illustrating polarization and spatial diversity strategies in compact 5G mmWave MIMO antennas. (**a**) Reflection coefficients for different antenna orientations in [84]. (**b**) Comparison between simulated and measured S_{11} and S_{12} results in [80]. (**c**) Effect of SRR loading on the dual-band response in [66].

In addition, several works combine diversity techniques with miniaturization and decoupling methods, including DGS and conformal structures, to simultaneously achieve compactness and high isolation [80,82]. The trade-off between ultra-compact size, wide bandwidth, stable radiation properties, and manufacturing complexity still presents significant hurdles, especially for wearable and multiband mmWave applications [66,82].

Table 5 compares representative compact mmWave MIMO antennas using polarization and spatial diversity techniques. The reported results indicate that these approaches can provide effective isolation and high radiation efficiency while preserving compact multi-port configurations. Most of the reviewed designs achieve isolation levels above 25 dB, while [80] reports a lower but still acceptable isolation range of 15–20 dB. This suggests that the effectiveness of polarization and spatial diversity depends strongly on the antenna arrangement, element orientation, and additional decoupling mechanisms used in the design.

From a quantitative point of view, the highest gain among the listed examples is reported in [128], reaching 8.9 dBi with an isolation higher than 29.24 dB and a radiation efficiency of 98.4%. The highest efficiency values are reported in [84], reaching 98.46% and 99.17% at the two operating bands, with isolation values of 30 dB and 25 dB. These results indicate that diversity-oriented antenna arrangements can maintain high radiation efficiency while reducing inter-port coupling.

In terms of frequency coverage, ref. [82] provides the widest operating range, from 26.5 to 40 GHz, with isolation higher than 50 dB. However, this result is reported as simulated, and the structure combines a conformal configuration, star-shaped slots, and DGS. Therefore, additional experimental validation would be useful to confirm its measured performance and practical robustness. By contrast, ref. [66] provides measured/simulated dual-band operation at 28/38 GHz with isolation higher than 25 dB and efficiencies of 92% and 89%, showing a more direct example of dual-band compact MIMO implementation.

Table 5. Summary of MIMO antennas employing polarization and spatial diversity techniques for 5G mmWave applications.

Ref.	Data Type	Freq. (GHz)	Ports	Substrate	Gain (dBi)	Bandwidth (GHz)	Isolation (dB)	Efficiency (%)	Diversity Method
[80]	Measured/ Simulated	28	4	Rogers RT/5880 (semi-flexible)	7.1	3.52	>15–20	>90	Dual DGS combined with polarization diversity
[84]	Measured/ Simulated	27.61–28.49/ 33.61–34.27	2	Duroid 5880	8.75/5.5	0.88/0.66	30/25	98.46/99.17	Slot-ring structure optimized via Taguchi neural network with polarization and diversity gain
[82]	Simulated	26.5–40	4	Rogers RT/ Duroid 5880	7.17	46.34	<-50	–	Conformal configuration with star-shaped slot and DGS enabling spatial and polarization diversity
[128]	Measured/ Simulated	25.785–29.485	4	Rogers RT/ Duroid 5880	8.9	3.70	>29.24	98.4	Machine-learning-assisted design (CST + ADS) for ECC reduction and diversity enhancement
[66]	Measured/ Simulated	28/38	4	Rogers RT5880 ($\varepsilon_r = 2.2$)	8.4/6.02	2.4/2	>25	92/89	Orthogonal element arrangement combined with CSRR to achieve spatial diversity

Overall, the comparison shows that polarization and spatial diversity are effective strategies for reducing correlation and improving isolation in compact mmWave MIMO antennas. Nevertheless, their performance should be evaluated together with other metrics such as gain, bandwidth, efficiency, number of ports, antenna geometry, additional decoupling structures, and validation type. Therefore, the best design is not necessarily the one with the highest isolation alone, but the one offering the most balanced compromise between diversity performance, radiation efficiency, bandwidth, compactness, and practical validation.

4.1.4. Integration of Special Materials

The integration of special materials, especially metamaterial-based structures, CSRRs, has recently gained considerable attention in the design of compact millimeter-wave MIMO antennas. These artificial materials, characterized by engineered electromagnetic properties, provide an effective means to mitigate mutual coupling while enhancing overall antenna performance [129–131]. As a result, metamaterial-inspired solutions represent a promising pathway to address the fundamental challenges of miniaturization, high isolation, and efficiency in millimeter-wave MIMO systems.

Metamaterials with electromagnetic properties not found in natural materials, such as negative permittivity or near-zero refractive index. These properties allow precise control of wave propagation, surface–wave suppression, impedance matching, and field manipulation, making them highly suitable for compact mmWave MIMO antennas [132].

Physically, metamaterial-based decoupling structures reduce mutual coupling by modifying the propagation of electromagnetic fields between adjacent antenna elements. Their periodic or resonant unit cells can introduce a stop-band response around the operating frequency, preventing surface waves and coupled currents from propagating across the antenna substrate. In addition, epsilon-negative, split-ring, or slab-based metamaterial structures can modify the local surface impedance and redirect the near-field energy away from the neighboring port. This explains why metamaterial decoupling cells are often inserted between MIMO elements or combined with modified ground planes to suppress dominant coupling paths and improve isolation [101,130,132].

Various designs in recent studies employ metamaterial-based decoupling structures, including epsilon-negative arrays, slabs, or optimized decoupling cells, often combined with modified ground planes or optimized element spacing [90,92,129,133,134]. Representative examples are shown in Figure 11. Low ECC, enhanced gain, good diversity

performance, and consistent efficiency across several bands are all attained by these methods. Metamaterial integration can hinder extreme downsizing for smartphone applications by increasing design complexity, multiband problems, and antenna area, even while it improves isolation and MIMO metrics.

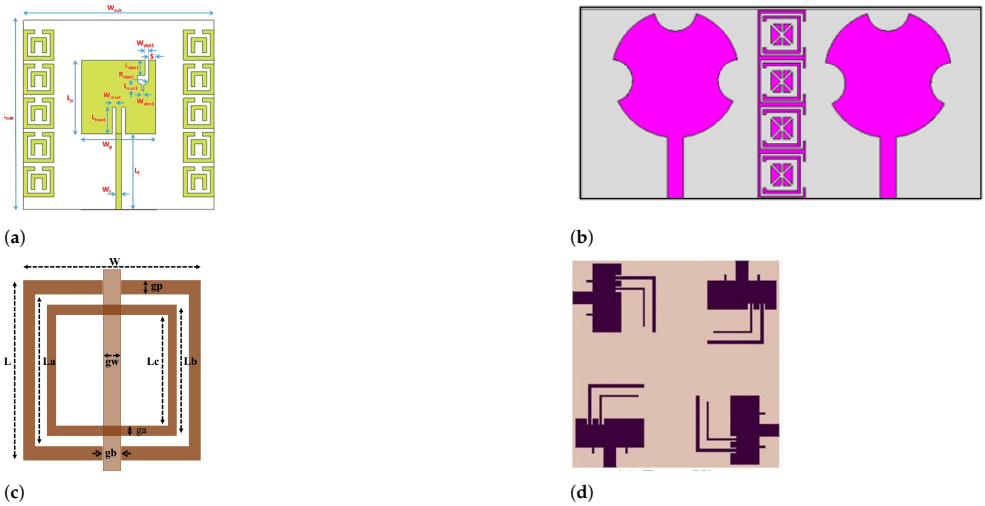


Figure 11. Examples of metamaterial-based and resonant structures used to improve isolation and diversity performance in compact millimeter-wave MIMO antennas. (**a**) Metamaterial-inspired antenna structure proposed in [92]. (**b**) MIMO configuration with metamaterial-based decoupling elements in [133]. (**c**) Metamaterial resonator structure used for dual-band MIMO performance enhancement in [129]. (**d**) Metamaterial-based decoupling structure integrated between MIMO elements in [90].

Complementary split-ring resonators (CSRRs), on the other hand, consist of periodic metallic rings etched on the ground plane, acting as band-stop filters that block surface currents. CSRRs improve isolation and radiation efficiency in dense arrays while maintaining compact size [101,135].

At the physical level, a CSRR behaves as an LC resonant element etched in the metallic ground plane or radiating surface. Around its resonant frequency, the CSRR produces a strong localized electromagnetic response that blocks or attenuates the surface current flowing between adjacent antenna elements. This band-stop behavior reduces the coupling path through the common ground plane and helps isolate the ports. The ring shape, slot width, gap position, and unit-cell placement determine the equivalent inductance and capacitance of the CSRR, and therefore control the resonant frequency and the strength of coupling suppression [101,135,136].

According to recent research, CSRR integration successfully improves isolation in mmWave MIMO antennas while preserving compactness, which makes it appealing for 5G applications [136–138]. Designs such as 2 × 2 printed patch antennas and slotted CSRR microstrip patches demonstrate that CSRR unit cells, etched on radiating surfaces and/or ground planes, control surface currents, reduce near-field coupling, and improve isolation. Modal-level analysis (TCM) provides additional insight into decoupling mechanisms and performance optimization; while CSRR-based techniques offer high gain, efficiency, and miniaturization potential, they are limited by array size constraints, increased design and optimization complexity, and, in some cases, restrictions for highly space-constrained or dense MIMO configurations.

Despite the proven effectiveness of complementary split-ring resonator (CSRR) structures for mutual coupling reduction and footprint miniaturization, their application to compact multi-port MIMO antennas at millimeter-wave frequencies remains relatively underexplored, with only a limited number of studies reported in the recent literature.

Table 6 compares representative compact MIMO antennas using metamaterial- and CSRR-based techniques. The reported results show that these approaches can improve isolation, gain, ECC, and efficiency in compact mmWave MIMO configurations. However, the performance varies significantly depending on the type of resonant structure used, the number of ports, the operating frequency, and the validation method.

Table 6. Comparative summary of special materials using metamaterials and CSRRs.

Ref.	Data Type	Frequency (GHz)	Ports	Gain (dBi)	Bandwidth (GHz)	Isolation (dB)	ECC	Efficiency (%)	Material/Technique Used
[129]	Measured/ Simulated	28/38	2	7.8 /6	4.8/4.2	20/12	<0.0001	—	Metamaterial (modified SRR) with ground plane modification for coupling reduction
[90]	Measured/ Simulated	3.5/5.2/28	4	4.8/5.3/9.3	10.01%/6.4%/ 4.43%	24/26/32	<0.002/0.001/ 0.0003	91	Epsilon-negative MTM array with partial ground and multi-stubbed patch; ML-assisted verification
[92]	Measured/ Simulated	28/35/38	4	9.6/7.8/13.7	1.1/2.2/3.7	34.5/22/30	0.00015	—	MTM-based decoupling cells with optimized element spacing for high-isolation MIMO
[133]	Measured/ Simulated	22–50	4	—	17	>30	0.0005	94	Metamaterial slab inserted between orthogonally oriented MIMO elements for high isolation and low coupling
[134]	Measured/ Simulated	27–29	2	8.75	2	64	<0.005	88	SRR metamaterial array between MIMO elements acting as band-stop filter and decoupler for high isolation
[137]	Measured/ Simulated	28/38	4	8.65/8.24	25.91–30.22/ 35.46–40.45	24/38.71	<0.0041	82.42	CSRR metamaterial unit cells on top and ground plane for improved gain, bandwidth, and isolation
[136]	Measured/ Simulated	27.4–28.5	2	8.31	1.1	27	0.001	89.5	Slotted CSRR metamaterial unit cell with TCM-based placement for high isolation and low coupling
[138]	Measured/ Simulated	28	4	13.7	27.39–29	—	—	97	CSRR resonators integrated on patch layer for miniaturization, high gain, and high efficiency

From a quantitative point of view, the strongest isolation is reported in [134], where an SRR metamaterial array inserted between the MIMO elements achieves 64 dB isolation at 27–29 GHz, with a gain of 8.75 dBi, ECC below 0.005, and 88% efficiency. The widest frequency coverage is reported in [133], covering 22–50 GHz with 17 GHz bandwidth, isolation higher than 30 dB, ECC of 0.0005, and 94% efficiency. These results indicate that metamaterial slabs and SRR-based decoupling arrays can provide strong coupling reduction while maintaining good diversity performance.

In terms of gain, the highest values are reported in [92,138], both reaching 13.7 dBi. The design in [92] operates at 28/35/38 GHz with isolation values of 34.5/22/30 dB and ECC of 0.00015, while [138] reports high gain and 97% efficiency at 28 GHz using CSRR resonators integrated on the patch layer. However, the isolation and ECC values are not reported in [138], which limits the completeness of the MIMO performance comparison.

CSRR-based designs also show balanced performance for dual-band mmWave MIMO operation. For example, ref. [137] achieves gains of 8.65/8.24 dBi at 28/38 GHz, isolation values of 24/38.71 dB, ECC below 0.0041, and efficiency of 82.42%. Similarly, ref. [136] reports 27 dB isolation, ECC of 0.001, gain of 8.31 dBi, and 89.5% efficiency around 28 GHz. These results confirm that CSRR unit cells can support coupling reduction and diversity enhancement while preserving compact antenna configurations.

Overall, metamaterial- and CSRR-based techniques offer strong potential for compact mmWave MIMO antennas, especially when high isolation, low ECC, gain enhancement, or miniaturization are required. Nevertheless, the comparison also shows that the best-

performing design depends on the targeted metric. SRR-based arrays provide the highest isolation, metamaterial slabs provide wide frequency coverage, and CSRR-based designs offer balanced dual-band operation. Therefore, these techniques should be evaluated not only by their maximum isolation or gain, but also by bandwidth, efficiency, number of ports, reported ECC, structural complexity, and experimental validation.

4.2. Structural and Feeding-Based Performance Optimization Techniques

4.2.1. Miniaturization Techniques

In the context of 5G and 6G millimeter-wave communications, the miniaturization of MIMO antennas aims to reduce the physical size of antenna modules to facilitate their integration into compact user devices and base stations, while preserving acceptable electromagnetic performance in terms of gain, radiation efficiency, and bandwidth [139]. Rather than representing a single design solution, antenna miniaturization encompasses a set of complementary approaches acting at different physical levels, including geometrical modification, material-based strategies, and the management of inherent performance trade-offs [140].

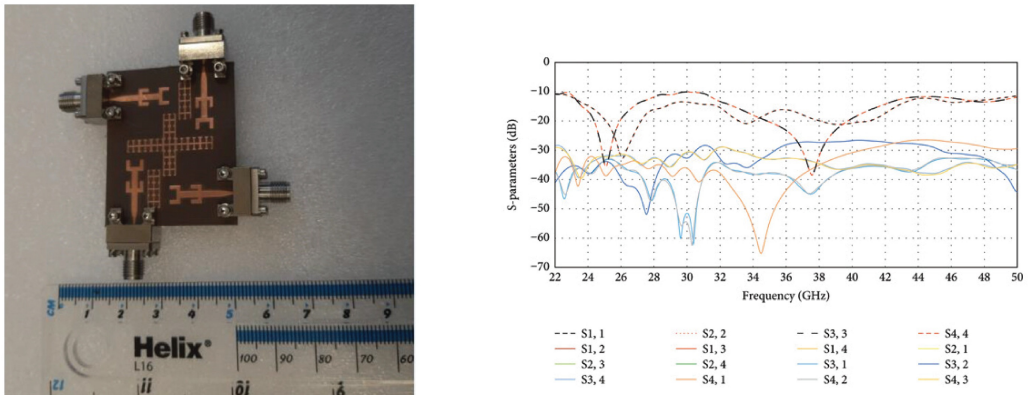
Geometrical miniaturization represents one of the most widely adopted approaches for reducing antenna size. This category relies on modifying the physical geometry of the radiating elements or the ground plane in order to increase the effective electrical length of the current path without enlarging the antenna footprint [141]. Techniques such as bending, folding, meandering, and the introduction of slots enable antennas to resonate at the desired frequencies within compact layouts, making them particularly suitable for space-constrained MIMO configurations [142].

Several recent studies have demonstrated the effectiveness of geometrical miniaturization. For instance, Sharma et al. [142] proposed a compact four-port MIMO antenna for 5G NR-FR2 applications, where optimized radiating geometries and element arrangements enabled triband operation while maintaining high isolation and stable radiation performance. Similarly, ref. [143] proposed a compact bent microstrip-based wideband MIMO antenna that employs a T-shaped strip with bent and L-shaped microstrip lines to reduce the antenna size and enhance performance.

Despite these advantages, geometrical miniaturization techniques may introduce fabrication complexity and impedance-matching challenges, particularly when three-dimensional configurations are involved, necessitating careful electromagnetic optimization during the design process [141].

Material-based miniaturization constitutes an alternative strategy, in which antenna size reduction is achieved by engineering the electromagnetic properties of the substrate or surrounding structures. The use of high-permittivity dielectric materials reduces the guided wavelength, allowing antennas to operate at a given frequency with smaller physical dimensions [144]. In addition, metamaterial-based structures have emerged as effective solutions for compact mmWave MIMO antennas, enabling both size reduction and enhanced control over electromagnetic fields [129,133]. For example, Din et al. [133] demonstrated a compact four-element mmWave MIMO antenna incorporating a metamaterial slab, achieving wideband operation with high isolation and radiation efficiency.

Figure 12 illustrates a representative example of material-based miniaturization. The fabricated prototype confirms the feasibility of integrating several radiating elements within a reduced antenna footprint, while the S-parameter results show wideband operation over the 22–50 GHz range with good impedance matching and high isolation between the antenna ports. This example supports the comparison presented in Table 7, where metamaterial-based structures are shown to improve compactness and isolation in dense mmWave MIMO configurations [133].



(**a**) (**b**)

Figure 12. Example of a compact metamaterial−based mmWave MIMO antenna and its S−parameter performance, adapted from [133]. (**a**) Fabricated prototype of the compact four port MIMO antenna. (**b**) Simulated and measured S−parameters.

However, material-based miniaturization inherently introduces important performance trade-offs. High-permittivity substrates and inclusions of metamaterials tend to increase the antenna quality factor, which may result in a reduced impedance bandwidth and increased dielectric losses [129,140].

Consequently, antenna miniaturization must be regarded as a multi-objective optimization problem, where compactness, bandwidth, radiation efficiency, and isolation must be jointly considered. Effective miniaturization strategies, therefore, require a careful balance between physical size reduction and acceptable electromagnetic performance, especially in dense multi-port antenna configurations [140].

Table 7. Comparison of representative miniaturization techniques in mmWave MIMO antennas.

Ref.	Data Type	Ports	Freq. Band (GHz)	Size (mm^2)	Bandwidth (GHz)	Gain (dBi)	Efficiency (%)	Miniaturization Approach
[145]	Measured/ Simulated	4	24.25–29.5	16.2 × 16.2	5.25	6.09	-	Geometrical miniaturization; circular patch, parasitic patch, vertical stubs; ultra-thin Rogers RT/Duroid 5880
[133]	Measured/ Simulated	4	22–50	36 × 28	28	-	>80	Material-based miniaturization; fork-shaped MIMO, metamaterial slab, partial ground planes; high isolation (65 dB)
[129]	Measured/ Simulated	2	28/38	3.8 × 3.7	3.8–4.8/ 3.8–4.2	7.8/6	-	Dual-band MIMO; embedded metamaterial unit cells; ground plane modification; low ECC, high DG
[143]	Measured/ Simulated	4	24.37–39.44/ 45.09–50.62	20 × 20	15.07/5.53	4.85–6.9	92–96	Bent microstrip lines; T- and L-shaped strips; fan-shaped decoupling; partial defective ground; low ECC, high DG
[146]	Measured/ Simulated	4	24–50	10.8 × 9	26	10.35	90	Compact four-port MIMO antenna for future 6G mmWave systems; metasurface-based isolation enhancement; FSS integration for gain improvement

Table 7 compares representative miniaturization strategies for compact mmWave MIMO antennas. The reported results show that miniaturization can be achieved through geometrical modification, material-based approaches, or hybrid combinations involving decoupling structures, metamaterials, metasurfaces, and FSS-based enhancement. How-

ever, the comparison also shows that antenna size reduction must be evaluated together with bandwidth, gain, efficiency, number of ports, and validation type.

From a quantitative point of view, the smallest antenna footprint is reported in [129], with a size of 3.8×3.7 mm^2 for a two-port dual-band MIMO antenna operating at 28/38 GHz. However, this design uses only two ports, while most other studies in the table use four-port configurations. Among the four-port designs, ref. [146] provides the most compact footprint, with a size of 10.8×9 mm^2, while covering the 24–50 GHz range with a bandwidth of 26 GHz, a gain of 10.35 dBi, and 90% efficiency.

In terms of bandwidth, ref. [133] provides the widest reported coverage, from 22 to 50 GHz, corresponding to a 28 GHz bandwidth, with efficiency higher than 80%. Similarly, ref. [146] covers 24–50 GHz with a 26 GHz bandwidth, while maintaining higher gain and a smaller antenna size. These results indicate that hybrid miniaturization strategies combining compact geometries with metamaterial or metasurface-based techniques can support wideband operation in dense mmWave MIMO configurations.

Geometrical miniaturization also provides a practical compromise between compactness and performance. For example, ref. [145] reports a compact four-port antenna of 16.2×16.2 mm^2 with 5.25 GHz bandwidth and 6.09 dBi gain, while [143] achieves dual-wideband operation with a 20×20 mm^2 four-port antenna, gain ranging from 4.85 to 6.9 dBi, and efficiency between 92% and 96%. These results show that bent lines, slots, parasitic patches, and partial ground modifications can reduce the antenna footprint while maintaining acceptable radiation performance.

Overall, the comparison confirms that miniaturization cannot be evaluated only by the smallest physical size. A highly compact design may involve fewer ports, narrower bandwidth, or additional metamaterial loading, while larger designs may provide wider bandwidth, higher gain, or better efficiency. Therefore, the most suitable miniaturization technique should be selected according to the required balance between antenna footprint, number of ports, bandwidth, gain, radiation efficiency, structural complexity, and experimental validation.

4.2.2. Bandwidth and Gain Enhancement

Improving bandwidth and gain is a major challenge and a key objective for antennas designed for 5G millimeter-wave systems. However, compact antenna structures such as microstrip patch or dipole antennas often suffer from narrow bandwidth and limited gain, which restricts their performance in high-frequency environments where propagation losses are significant. Therefore, enhancing bandwidth and gain while maintaining compact dimensions is essential to compensate for millimeter-wave attenuation and to improve radiation efficiency, directivity, and link stability in 5G networks [61,147].

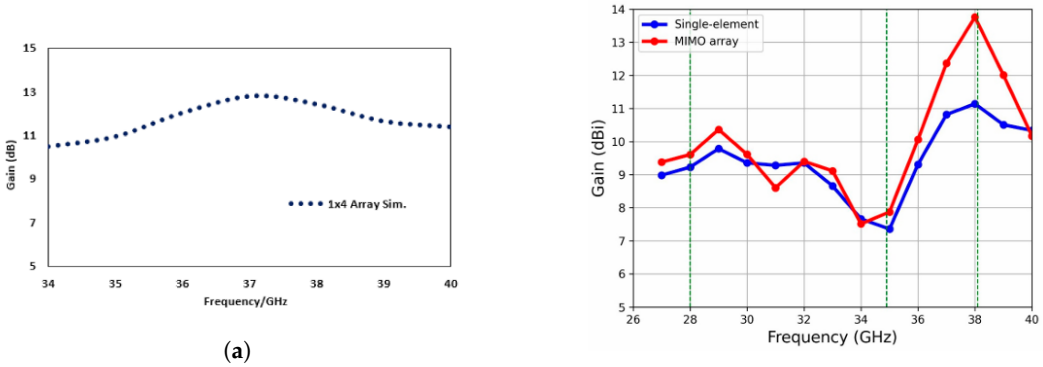
Wideband antenna design strategies: The design of wideband antennas remains a key requirement in 5G millimeter-wave systems, where large bandwidth is necessary to support multi-gigabit data rates, beamforming, and MIMO operations. Several geometric strategies have been explored to extend impedance bandwidth while preserving compactness and radiation performance.

Geometric optimization techniques, including slotted, tapered, and stacked configurations, play a central role in bandwidth enhancement. For instance, ref. [148] proposed a compact antipodal Vivaldi antenna operating over the Ka-band (23–40 GHz). The exponential taper enables wideband behavior and high directivity, while the integration of Epsilon-Near-Zero (ENZ) metamaterials enhances electromagnetic field confinement and improves gain without degrading bandwidth. However, the proposed structure increases overall complexity and antenna size and is limited to a single-element configuration without MIMO evaluation.

Gain-enhancement approaches: Gain enhancement has become a critical design requirement for compact MIMO antennas intended for advanced applications. Traditionally, antenna arrays increase gain by enlarging the effective aperture, but at the cost of increased footprint, feeding complexity, and fabrication cost. To overcome these limitations while preserving compactness, several alternative gain-enhancement techniques have been proposed in recent literature. These include array configurations, parasitic elements, metamaterial-based solutions, and superstrate loading [149].

Several strategies have been reported. For instance, ref. [92] presents a high-performance four-port MIMO antenna operating at multiple 5G mmWave bands (28, 35, and 38 GHz), achieving peak gains of up to 13.7 dBi in the tri-band configuration. Improved inter-element spacing and the use of metamaterial cells for decoupling significantly contributed to both isolation and gain enhancement, while maintaining low envelope correlation and high diversity gain. Similarly, a compact MIMO antenna system has demonstrated that introducing a dielectric layer behind the radiating elements can increase the peak gain from approximately 5.48 dB to 8.47 dB while preserving wide bandwidth and high efficiency [150]. An alternative approach employs array configurations to improve the overall MIMO system gain. In [34], a four-element MIMO antenna array operating at 37 GHz achieved a significant gain enhancement (up to 12.8 dB) compared to a single element, without significantly degrading efficiency. This demonstrates that array techniques, when carefully optimized in terms of element spacing and feeding structure, remain effective gain-boosting mechanisms in mmWave MIMO systems.

As illustrated in Figure 13, the use of array-based MIMO configurations leads to a clear improvement in the antenna gain compared with single-element structures. In Figure 13a, the simulated 1×4 array shows a stable gain response over the 34–40 GHz frequency range, with a maximum value close to 12.8 dBi. This confirms that increasing the number of radiating elements can enlarge the effective aperture and improve the radiation performance of the antenna system.



(**a**) (**b**)

Figure 13. Gain-enhancement behavior of mmWave MIMO antenna arrays. (**a**) Simulated gain of the 1×4 antenna array reported in [34]. (**b**) Gain comparison between a single element and the MIMO array reported in [92]. The green dashed vertical lines in subfigure (**b**) indicate the selected operating frequencies at 28, 35, and 38 GHz.

Similarly, Figure 13b shows that the MIMO array achieves higher gain than the single antenna element over most of the considered mmWave band. The gain enhancement is particularly noticeable, where the MIMO array reaches a peak gain of approximately 13.7 dBi. This behavior demonstrates that a carefully optimized MIMO arrangement can significantly improve gain while maintaining operation over multiple mmWave bands.

Impedance-matching techniques: Impedance matching plays a fundamental role in maximizing power transfer between the feed network and the radiating elements in

millimeter-wave antenna systems. Optimized matching reduces reflection losses and improves overall radiation efficiency.

In this context, ref. [151] conducted a comparative study of several matching techniques for microstrip antenna arrays operating around 28 GHz. The investigated approaches included quarter-wave transformers, single-stub tuning circuits, and smooth corner modifications of the patch geometry. These methods significantly reduced the reflection coefficient, achieving an S_{11} value of −75 dB at 28 GHz and a bandwidth of 8.3%. Among them, the smooth corner configuration provided effective impedance matching and a high gain of 13 dB while maintaining a low-profile planar structure. Similarly, ref. [152] proposed stacking a parasitic patch above the main inset-fed radiating element, separated by an optimized air gap. This coupling mechanism introduced additional resonances that extended the impedance bandwidth across the 25–31 GHz range, leading to improved return loss, gain, and radiation efficiency compared to conventional single-patch designs.

Table 8 compares representative mmWave antenna designs used for bandwidth and gain enhancement. Since the listed works do not all have the same configuration, including single-element antennas, antenna arrays, and MIMO systems, the comparison should be interpreted according to the number of ports, reported isolation, ECC, and validation type.

Table 8. Performance comparison of representative mmWave antennas for bandwidth and gain enhancement.

Ref.	Data Type	Freq. (GHz)	Substrate	Ports	Bandwidth	Gain (dBi)	Isolation (dB)	ECC	Efficiency (%)
[148]	Measured/ Simulated	23–40	Rogers RO4003C	1	17 GHz	17.2	–	–	–
[92]	Measured/ Simulated	28/35/38	Rogers RT-5880	4	1.1/2.2/3.7 GHz	13.7	22–34.5	0.00015	–
[150]	Measured/ Simulated	16.7–25.4	Rogers RO-4350B	2	8.7 GHz	8.47	>30	<0.5	>80
[34]	Measured/ Simulated	37	Rogers RT5880	8 (2 × 4)	1.35	12.8	>40	<0.00014	>85
[151]	Simulated	28	Duroid 5880	4 (1 × 4)	8.3%	13	–	–	–
[152]	Measured/ Simulated	24.35–31.13	Rogers RT5880	16	(6.78) 24.4%	19.88	20	–	>86

From a bandwidth perspective, the widest operating range is reported in [148], covering 23–40 GHz with a 17 GHz bandwidth and a gain of 17.2 dBi. However, this design corresponds to a single-element antenna and does not provide MIMO-related metrics such as isolation or ECC. Among the MIMO and array-based configurations, ref. [150] reports an 8.7 GHz bandwidth over the 16.7–25.4 GHz range, with isolation higher than 30 dB and efficiency above 80%. Similarly, ref. [152] achieves a 6.78 GHz bandwidth, high gain of 19.88 dBi, isolation of 20 dB, and efficiency higher than 86%.

In terms of gain, the highest value is reported in [152], reaching 19.88 dBi with a 16-port array configuration. The designs in [34,92,148,151] also report high gain values of 17.2 dBi, 13.7 dBi, 12.8 dBi, and 13 dBi, respectively. These results show that gain enhancement is mainly achieved through wideband radiating structures, array configurations, metamaterial-based decoupling, dielectric loading, parasitic patches, or impedance-matching techniques.

Regarding MIMO performance, ref. [34] provides the strongest isolation among the listed MIMO designs, with isolation higher than 40 dB, ECC below 0.00014, gain of 12.8 dBi, and efficiency above 85%. The design in [92] also provides a balanced tri-band response at 28/35/38 GHz, with gain up to 13.7 dBi, isolation between 22 and 34.5 dB, and ECC of 0.00015. These results indicate that gain enhancement must be evaluated together with isolation and ECC when the objective is compact MIMO operation.

Overall, the comparison shows that wide bandwidth and high gain can be achieved using different strategies, but the best-performing design depends on the intended objective. Single-element wideband antennas can provide large bandwidth and high gain, while MIMO and array-based configurations must also satisfy isolation, ECC, efficiency, and multi-port integration requirements. Therefore, bandwidth and gain-enhancement techniques should be evaluated through a combined analysis of frequency coverage, gain, number of ports, isolation, ECC, efficiency, and validation type.

4.2.3. Feeding Techniques and Antenna Arrays

In millimeter-wave (mmWave) MIMO systems, the feed network and array configuration are fundamental to achieving high gain, effective beamforming, and reliable multi-element operation. Proper feeding ensures uniform phase and amplitude distribution among radiating elements, which is critical for accurate beam steering, minimized mutual coupling, and compensation of propagation losses [153]. The most common feeding architectures include series-fed, parallel-fed, hybrid-fed, and corporate-fed networks, each offering distinct advantages and trade-offs.

Series-fed arrays are compact and efficient structures in which radiating elements are sequentially connected along a single transmission line. This configuration reduces feeding complexity and substrate area, making it suitable for compact mmWave arrays. However, conventional series-fed designs may suffer from unequal power distribution and limited bandwidth [154–156].

In recent studies, several innovations have been proposed. The first article [157] proposed a four-port MIMO antenna operating at 28 GHz, where each port excites a configuration of the 1×8 series-fed array. The series-feeding technique significantly enhances gain (up to 15.5 dBi) while maintaining a 2 GHz bandwidth. Through optimized element spacing and metamaterial-based decoupling cells, the effectiveness of series-fed architectures for high-performance mmWave MIMO systems is highlighted.

The corporate feed network can ensure uniform amplitude and phase across multiple elements. Implementations have demonstrated high gains, stable radiation patterns, and effective beam-steering capabilities. Another article [158] presents a reconfigurable beam-steering antenna array, using a corporate-fed power divider and a phase-shifting mechanism by implementing a 3-bit phase shift across eight feed points. However, the complexity of the switching mechanism and the need for precise phase alignment may limit its practicality in real-world implementations.

Aperture-coupled feeding, another widely used approach in mmWave arrays, transfers energy from the feed line to the patch via a slot in the ground plane rather than direct contact. This indirect coupling allows independent optimization of the feed and radiating structures, improving impedance matching, reducing back radiation, and improving isolation between elements. For example, ref. [159] proposes a dual-polarized aperture-coupled antenna operating between 24 GHz and 44 GHz. The low complexity and cost-effective nature of the antenna, combined with polarization discrimination exceeding 20 dB, present it as an ideal candidate for massive MIMO systems. However, the simple structure may limit performance in high-density applications requiring additional enhancements. Another example that [160] proposed is a planar multi-circular loop antenna operating from 26.5 to 41 GHz. The single element is transformed into a four-element array with a compact footprint (18.25×12.5 mm^2). Although the feeding structure is relatively simple, the array configuration enhances gain performance while maintaining a wide impedance bandwidth, making it suitable for next-generation mmWave systems.

As shown in Figure 14, the radiation patterns that were simulated and those that were observed agree well in both main planes. This agreement says that the suggested array configuration and feeding structure are valid.

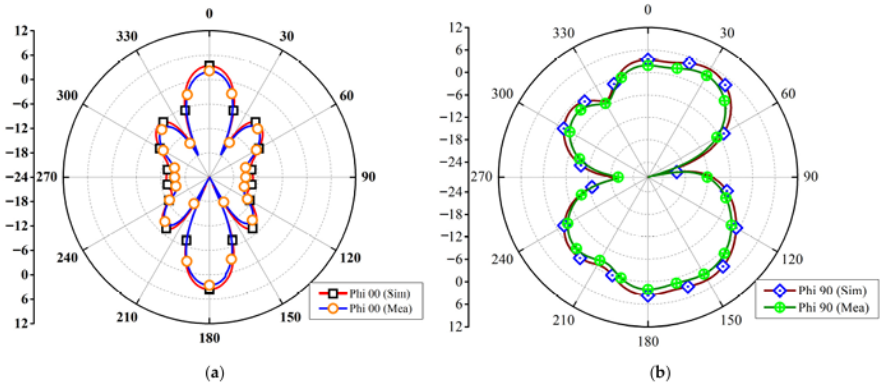


Figure 14. Simulated and measured radiation patterns of the proposed antenna array reported in [160]: (**a**) radiation patterns in the $\phi = 0^\circ$ plane; (**b**) radiation patterns in the $\phi = 90^\circ$ plane.

Beamforming and phased array technologies are essential for mmWave MIMO networks, enabling directional and energy-efficient transmission. Beam steering is achieved by controlling the phase and amplitude of each element's excitation, concentrating energy toward specific users, relays, or sectors, which improves signal-to-interference-plus-noise ratio (SINR) while mitigating mmWave propagation losses [161]. Beamforming can be implemented in analog, digital, or hybrid architectures. Analog beamforming uses RF phase shifters, offering low cost and energy efficiency but limited flexibility. Digital beamforming allows precise amplitude and phase control, supporting multiple simultaneous beams, yet requires a dedicated RF chain per element, increasing complexity and power consumption. Hybrid beamforming combines analog and digital processing to balance performance, energy efficiency, and complexity, making it particularly suitable for massive MIMO systems at mmWave frequencies [162,163].

Table 9 compares representative feeding techniques and antenna-array configurations used in mmWave antenna systems. The reported results show that the feeding architecture strongly affects gain, bandwidth, efficiency, beam-control capability, and MIMO performance. Therefore, feeding techniques should not be evaluated only in terms of gain, but also according to bandwidth, number of ports, isolation, ECC, efficiency, and implementation complexity.

Table 9. Comparison of feeding techniques and array performances in mmWave antenna systems.

Ref.	Data Type	Freq. (GHz)	Substrate	Ports	Bandwidth	Gain (dBi)	Isolation (dB)	ECC	Efficiency (%)	Feeding Technique
[158]	Measured/ Simulated	28	RT/duroid 5880	8	25.8 to 30.5	13.44	-	-	68 to 81	Corporate-fed Phase Shifter
[159]	Measured/ Simulated	24–44	Rogers RT Duroid 5880	2	60%	7	>35	-	-	Aperture-coupled feed
[160]	Measured/ Simulated	26–38.5	RO5880	1 × 4	15.5	11.5	-	-	95	Planar (Multi-Circular Loop)
[157]	Measured/ Simulated	28.8	Rogers RT-5880	4	2.0	15.5	>40	<0.00010	-	Series-fed
[164]	Measured/ Simulated	30.5	Rogers RT-5880	4	7.85	14.1	>40	<0.00010	36	Series-fed + Metamaterials

From a quantitative point of view, the highest gain is reported in [157], reaching 15.5 dBi at 28.8 GHz with a four-port series-fed architecture. This design also achieves

isolation higher than 40 dB and ECC below 0.00010, indicating strong MIMO performance. A similar series-fed approach combined with metamaterials is reported in [164], with 14.1 dBi gain, 7.85 GHz bandwidth, isolation higher than 40 dB, and ECC below 0.00010. However, its reported efficiency is 36%, showing that high gain and isolation should be evaluated together with radiation efficiency.

Corporate-fed and phase-shifting networks are more directly related to beam-steering and adaptive radiation control. In [158], the corporate-fed phase-shifter array operates around 28 GHz with eight ports, a frequency range from 25.8 to 30.5 GHz, gain of 13.44 dBi, and efficiency between 68% and 81%. Although isolation and ECC are not reported in the table, this configuration highlights the relevance of phase-controlled feeding networks for reconfigurable and beam-steerable mmWave antenna systems.

Aperture-coupled feeding also shows strong wideband potential. The design in [159] covers 24–44 GHz with 60% bandwidth, gain of 7 dBi, and isolation higher than 35 dB. This indicates that aperture-coupled structures can support broadband and dual-polarized mmWave operation. In addition, the planar multi-circular loop array in [160] reports 11.5 dBi gain and 95% efficiency, showing that array configuration can improve radiation performance while maintaining high efficiency.

Overall, the comparison shows that each feeding technique provides a different performance compromise. Series-fed arrays can achieve high gain and strong isolation with compact feeding structures, but their efficiency and bandwidth must be carefully checked. Corporate-fed phase-shifting networks are suitable for beam-steering and adaptive radiation control, but they require more complex phase and amplitude management. Aperture-coupled and planar array configurations can provide wideband behavior or high efficiency, depending on the design. Therefore, the most suitable feeding architecture should be selected according to the required balance between gain, bandwidth, isolation, ECC, efficiency, beamforming capability, number of ports, and implementation complexity.

4.3. Advanced Design Methodologies

4.3.1. AI and Machine-Learning-Based Optimization

The use of artificial intelligence (AI) and machine learning (ML) in the design and optimization of millimeter-wave (mmWave) MIMO antennas for 5G communications significantly reduces design time while improving overall performance quality [165]. ML and deep learning (DL) techniques leverage electromagnetic (EM) simulators such as CST and HFSS to build predictive models that accelerate the optimization process through methods such as multi-objective optimization, surrogate-based modeling, parallel optimization, and multifidelity optimization [166]. Evolutionary algorithms, including genetic algorithms (GA), particle swarm optimization (PSO), and differential evolution (DE), combined with supervised learning models, enhance convergence speed and robustness in achieving optimal antenna designs [165]. Other approaches incorporate domain knowledge or generative techniques to efficiently explore the design space and obtain reliable solutions under complex constraints [165]. Overall, these AI/ML-driven strategies enable accurate antenna behavior prediction, reduce the number of required EM simulations, and significantly accelerate the development of compact MIMO antennas for 5G and future 6G mmWave communication systems [166].

AI/ML-based antenna design methodologies can be broadly divided into several categories. The first category includes surrogate-based models, such as artificial neural networks, Gaussian process regression, support vector regression, and deep neural networks, which are trained to predict antenna responses such as reflection coefficient, gain, bandwidth, radiation efficiency, or isolation without repeatedly running full-wave simulations [165–168]. The second category includes evolutionary and population-based optimization algorithms, such as

GA, PSO, DE, and MOGA, which are used to search for optimal antenna geometries under multi-objective constraints [165,169]. The third category includes reinforcement-learning and automated-design frameworks, where the design parameters are iteratively updated according to a reward function related to antenna performance [170]. These categories show that advanced computational design is not a single technique, but a family of tools that can support prediction, optimization, automation, and design-space exploration [165,166].

In practical antenna design, AI/ML-based optimization is mainly used to accelerate simulation-driven design by replacing repeated and computationally expensive full-wave electromagnetic simulations with surrogate or predictive models [165,166]. These models are usually trained from datasets generated using electromagnetic simulators such as CST or HFSS, and can support antenna behavior prediction, design space exploration, and single- or multi-objective optimization [166].

Surrogate-assisted optimization is particularly useful for complex mmWave antenna and array designs, where repeated full-wave simulations are time-consuming. For example, Tulum et al. [171] showed that data-driven surrogate modeling can strongly reduce the computational time required for phased-array antenna optimization. More recently, Quan et al. [170] proposed an automated antenna design framework combining supervised learning and reinforcement learning to reduce dependence on manual tuning and expensive electromagnetic simulations.

For compact mmWave MIMO antennas, these approaches are promising because several objectives must be optimized at the same time, including impedance matching, bandwidth, gain, isolation, ECC, efficiency, and compactness [165,166]. However, AI/ML-based antenna optimization still depends strongly on the quality and diversity of the training data, and its physical interpretability remains limited compared with physics-based approaches [165,166]. Therefore, future research should combine AI/ML with electromagnetic insight, physics-based methods, and experimental validation to improve both optimization efficiency and practical reliability [165,166].

Numerous studies have demonstrated the growing impact of artificial intelligence (AI) and machine learning (ML) techniques. Genetic algorithms, including quantum and dual-phase variants, have been used to enhance power efficiency, beam selection, and bandwidth in massive MIMO and microstrip antenna designs [172,173]. More recently, Quan et al. [170] proposed an automated antenna design approach. The method demonstrates how reinforcement learning can be used to explore complex antenna design spaces and reduce reliance on manual parameter tuning. Such approaches are particularly promising for compact mmWave MIMO antennas, where several objectives, including isolation, gain, and compactness, must be optimized simultaneously.

Another example, ref. [169] designed a directional SIW horn antenna for mmWave 5G broadcasting and optimized its structural parameters using a multi-objective genetic algorithm (MOGA). The evolutionary optimization approach enabled improved directivity and gain performance. Although the design targets a directional antenna rather than a MIMO configuration, this work demonstrates the effectiveness of genetic-algorithm-based optimization for enhancing radiation characteristics in mmWave antenna systems. Another article [174] proposed a compact 4×4 microstrip patch array antenna operating in the 24–40 GHz band. The antenna achieves balanced performance and high efficiency, demonstrating the effectiveness of multi-element array configurations in enhancing radiation performance, the methodology used may limit the achievement of optimal performance and does not address coupling reduction or intelligent optimization strategies.

In parallel, ANN-based models have shown high accuracy in predicting antenna S-parameters and frequency responses, achieving over 200× faster computation compared to traditional EM simulations [167,168]. Several recent studies have demonstrated the

effectiveness of neural networks. The authors of [171] proposed a data-driven surrogate modeling approach for phased array antennas operating in the millimeter-wave band. The surrogate-assisted optimization framework achieved nearly 90% reduction in computational time compared to conventional full-wave EM simulations. Although the study focuses on phased array systems rather than compact MIMO antennas, it clearly demonstrates the potential of deep-learning-based surrogate models for accelerating the design and optimization of complex mmWave antenna systems involving strong coupling effects and multi-parameter optimization.

4.3.2. Characteristic Mode Analysis (CMA)

The Theory of Characteristic Modes (CMA) provides a systematic and physics-based framework for analyzing and designing antennas by decomposing the surface current distribution into orthogonal characteristic modes. These modes represent the natural resonant behaviors of the structure, each associated with a distinct radiation pattern. Understanding these radiation modes enables designers to perform mode-based design, where specific modes are selectively excited to achieve desired performance objectives such as improved gain, bandwidth, or polarization purity [175]. In MIMO systems, CMA is particularly useful for identifying and exciting orthogonal modes to reduce mutual coupling and achieve low envelope correlation with high diversity gain. By analyzing the modal significance and current distribution, designers can enhance radiation efficiency while suppressing unwanted coupled or non-radiating modes [176]. Overall, CMA serves as a powerful tool for understanding the fundamental radiation behavior of MIMO antennas and guiding mode-based design for next-generation 5G systems with optimized radiation and isolation characteristics [176].

Recent studies have begun to explore the application of Characteristic Mode Analysis (CMA) as a systematic and physics-based approach for the design and optimization of MIMO antennas, with emerging interest in millimeter-wave 5G communications; Abdelaziz et al. [175] applied the Theory (CMA) to systematically design a 5G logarithmic spiral MIMO patch antenna operating around 28 GHz. The objective was to exploit modal behavior to optimize antenna placement and achieve wide bandwidth, high isolation, and low ECC. However, its limitations lie in the restricted two-port MIMO configuration and the lack of a comparative analysis with alternative decoupling techniques, which limits the generalizability of the CMA-based optimization approach. Similarly, ref. [177] demonstrated the effectiveness of CMA in the design of a compact and highly flexible millimeter-wave antenna operating between 28 and 32 GHz. The study systematically analyzed the eigenvalues and modal currents through multiple design stages. Although the proposed structure is not a MIMO configuration, the work confirms that CMA provides an efficient tool for controlling modal excitation and bandwidth enhancement in compact mmWave antennas.

It is worth noting that the majority of existing CMA-based MIMO antenna designs are primarily reported in the sub-6 GHz frequency bands. Although CMA has proven highly effective for modal-based decoupling and diversity enhancement at lower frequencies, its application in compact millimeter-wave MIMO systems remains relatively limited. This highlights an active research gap in the literature, particularly for highly integrated mmWave 5G antenna configurations.

In practice, CMA-based antenna design usually follows a mode-oriented workflow. First, the characteristic modes of the antenna platform are calculated without excitation. Then, modal significance, characteristic angle, eigenvalues, and modal current distributions are analyzed to identify the modes that contribute effectively to radiation and those that may increase coupling or correlation [175,176]. Finally, feeding positions, slots, ground

modifications, or element orientations are selected to excite the desired orthogonal modes while suppressing undesired modes [175,177]. This makes CMA particularly useful for compact MIMO antennas, where the main challenge is not only to obtain impedance matching, but also to control modal orthogonality, reduce ECC, and improve isolation without relying only on empirical decoupling structures [175,176].

Table 10 compares representative advanced design methodologies used for mmWave antenna and MIMO antenna optimization, including AI/ML-based optimization and Characteristic Mode Analysis (CMA). Since the listed studies do not all have the same antenna configuration, the comparison should be interpreted according to the design methodology, number of ports, validation type, and reported MIMO metrics.

Table 10. Comparative analysis of advanced design methodologies for compact mmWave MIMO antennas.

Ref.	Data Type	Freq. (GHz)	Substrate	Ports	Bandwidth	Gain (dBi)	Isolation (dB)	ECC	Efficiency (%)
[169]	Simulated	28.3/38.1/46.6/60.0	RT/Duriod-5880	1	11.5	7.8/8.8/7.3/7.1	-	-	96.3/95.4/97.7/94.8
[174]	Simulated	24–40	Rogers RT/Duroid 5880	4	2.52	9.54	−26.71	-	86.82
[171]	Simulated	26–35	Rogers 4350B	4	30%	13	−20	-	-
[175]	Measured/Simulated	27.3–30.2	Rogers RT5880	2	−10	9.9	>−36	<0.002	94
[177]	Measured/Simulated	29–32.2	Polyimide	2	28.72–32	5.6	>−55	<0.002	90

From a quantitative point of view, AI/ML-based and evolutionary optimization approaches are mainly used to accelerate the design process and improve radiation performance. For example, ref. [169] reports a simulated multiband antenna operating at 28.3/38.1/46.6/60 GHz with radiation efficiencies higher than 95% at all bands. However, this design is a single-port antenna, so isolation and ECC are not reported. In contrast, refs. [171,174] present four-port configurations, with gains of 9.54 dBi and 13 dBi, respectively, showing the relevance of data-driven and optimization-based methods for multi-port mmWave antenna design.

CMA-based approaches provide a different advantage because they offer physical insight into the modal behavior of the antenna structure. In [175], CMA is applied to a two-port 28 GHz MIMO antenna, achieving a gain of 9.9 dBi, isolation higher than 36 dB, ECC below 0.002, and efficiency of 94%. Similarly, ref. [177] reports isolation higher than 55 dB, ECC below 0.002, and 90% efficiency over the 29–32.2 GHz range. These results indicate that CMA can support low-correlation and high-isolation antenna designs by identifying useful characteristic modes and controlling modal excitation.

Overall, the comparison shows that AI/ML-based methods and CMA are complementary rather than competing approaches. AI/ML techniques are useful for fast prediction, design-space exploration, and multi-objective optimization, especially when many geometrical parameters must be optimized. CMA, on the other hand, provides a physics-based understanding of current distributions, radiation modes, and coupling mechanisms. Therefore, future compact mmWave MIMO antenna design can benefit from hybrid strategies that combine the optimization speed of AI/ML with the electromagnetic interpretability of CMA.

5. Challenges and Limitations

5.1. Electromagnetic and Physical Constraints

At mmWave frequencies, compact MIMO antenna design is limited by high propagation losses, atmospheric attenuation, and short transmission range, requiring antennas with high gain and efficiency [14,178]. Miniaturization also increases mutual coupling between closely spaced elements, reducing diversity performance. Techniques such as EBG, metamaterials, or PDGS can suppress coupling and surface waves, but often add complexity or limit bandwidth [179–181]. The main challenge remains balancing compactness, isolation, and efficiency.

Following the "Challenges and limitations" branch of Figure 2, this section discusses the main electromagnetic, material, fabrication, integration, and manufacturing constraints that limit the practical implementation of compact mmWave MIMO antennas.

5.2. Substrate and Material Challenges

Substrate properties—permittivity, loss tangent, and thickness—directly affect gain, bandwidth, and efficiency of mmWave antennas. Low ε_r and low $\tan\delta$ improve efficiency but increase size, while thick substrates excite surface waves, increasing coupling [34,132]. Advanced substrates, such as metamaterial-based designs, enhance isolation without compromising compactness. As shown by [182], careful optimization of miniaturization, gain, and thermal stability is essential for 5G mmWave MIMO antennas. The next section compares rigid, flexible, and SIW-based substrates for practical 5G applications.

5.2.1. Rigid Substrates

Rigid substrates such as Rogers RO4003C, RO5880, and FR-4 are widely used in compact MIMO antennas for their stable dielectric properties and structural reliability. High-end materials like Rogers RO5880 ($\varepsilon_r = 2.2$, $\tan\delta \approx 0.0009$) and RO4003C ($\varepsilon_r = 3.55$, $\tan\delta \approx 0.0027$) provide low losses and high radiation efficiency, but are costly and difficult to process for large-scale or flexible integration [183,184]. Low-cost alternatives such as FR-4 ($\varepsilon_r \approx 4.4$, $\tan\delta \approx 0.02$) are easier to use but suffer from high losses, poor thermal stability, and reduced efficiency at mmWave frequencies [185]. Overall, rigid substrate selection involves a trade-off between cost, dielectric performance, and fabrication feasibility.

5.2.2. Flexible Substrates

Flexible substrates (polyimide, PDMS, textile, LCP) are attractive for wearable and IoT mmWave MIMO antennas due to conformability and lightweight nature. However, many exhibit high dielectric losses, moisture absorption, and thermal expansion issues, reducing efficiency, bandwidth, and isolation [13,186,187]. LCP and 3D-printed polymers like polypropylene offer low losses and better moisture resistance, but processing costs, adhesion, and surface roughness remain challenges [188,189]. Mechanical bending and on-body placement further affect resonant frequency and coupling, making reliable, low-loss operation under stress a key limitation [190,191].

5.2.3. Substrate-Integrated Waveguide (SIW) Structures

SIW structures offer planar integration, high isolation, and low profile, combining traditional waveguide benefits with PCB compatibility [192,193]. They suppress surface waves and mutual coupling, making them suitable for dense MIMO and phased-array systems, while enabling low-cost, scalable 5G/6G implementations. However, via misalignment, metallization variations, and surface roughness can detune resonances and reduce isolation [194]. Miniaturized cavities increase radiation loss and reduce bandwidth, and maintaining low insertion loss across multilayer or flexible substrates remains chal-

lenging [195]. Optimization in via design and hybrid integration can enhance robustness and scalability.

5.3. Fabrication and Integration Limitations

Tolerances and alignment: mmWave antennas require extremely tight fabrication tolerances; small deviations can shift resonant frequencies, alter matching, and degrade radiation patterns [12,196,197]. Even 25 µm size changes can shift frequency by hundreds of MHz, emphasizing the need for precise control in multilayer and high-density packaging [198].

Feeding network complexity: Feeding networks must maintain accurate phase and amplitude across elements for beamforming. Geometric deviations, phase imbalance, or mismatch can degrade MIMO performance [196,199]. Designing robust networks tolerant to minor errors is critical.

Integration with RF front-end: Antenna integration with filters, dividers, and connectors can introduce coupling, thermal stress, and parasitic effects [12,140]. AiP and chip-level solutions help compact designs, but misalignments or substrate variations affect efficiency and impedance matching [196,197]. Ensuring reliable transitions and repeatable manufacturing remains a key limitation for mass production of mmWave modules.

5.4. Emerging Substrate Technologies

5.4.1. Multilayer and Heterogeneous Substrates

Multilayer and heterogeneous substrates enhance compactness, bandwidth, and integration in mmWave antennas [12,200–202]. Screen-printed polymer–ceramic multiple layers enable low-loss, stacked patch antennas with stable performance under bending, suitable for flexible applications [200]. Dimensional variations, such as prepreg or core thickness deviations, can shift resonant frequencies by 130–280 MHz, highlighting sensitivity to manufacturing tolerances [201]. Materials like PTFE, LCP, hydrocarbon laminates, microwave ceramics, and LTCC provide multilayer integration, cross-layer vias, and embedded cavities, supporting compact planar antennas from mmWave to sub-mmWave frequencies [12,202].

5.4.2. Three-Dimensional-Printed and Additively Manufactured Substrates

Additive manufacturing and printing processes have gained attention as alternative fabrication methods for mmWave antennas. developed a fully screen-printed additive process, where both the dielectric and metallic layers were printed using custom dielectric ink and silver nanowire (AgNW) conductive ink. This approach enabled low-cost, fully printed multilayer antennas suitable for flexible applications [200].

Wagih et al. [203] reviewed mmWave power-harvesting systems and noted that additively manufactured circuits and packages are increasingly used for low-cost fabrication of antennas and rectifiers. They highlighted implementations on polymer and textile substrates, indicating the compatibility of additive manufacturing with flexible and wearable systems. Yang et al. [12] also mentioned 3D printing and MEMS-based fabrication processes as emerging methods for mmWave antenna realization, complementing conventional LTCC and ceramic technologies.

5.4.3. Metamaterial-Based Substrates

In [202], presented metamaterial-based antennas at 60 GHz, including a zero-index metamaterial (ZIM) structure and a mushroom-type antenna array. The ZIM-loaded antenna achieved a gain enhancement of up to 2.6 dB between 57 and 66 GHz, demonstrating the benefit of metamaterial integration in mmWave substrate designs. They also reported

that such metamaterial configurations can be implemented on LTCC and PCB substrates, combining metamaterial effects with substrate-integrated waveguide (SIW) techniques.

Although metamaterial-based and multilayer antenna structures can improve compactness, isolation, gain, and bandwidth, their practical manufacturability remains challenging at mmWave and sub-THz frequencies [12,140,204]. Their performance strongly depends on the accurate realization of small resonant elements, vias, slots, unit cells, and interlayer transitions. At high frequencies, even small dimensional deviations may shift the operating frequency and degrade impedance matching, isolation, radiation efficiency, and repeatability [12,204].

In multilayer and package-integrated antennas, additional reliability issues can arise from layer misalignment, substrate-property variations, via/interconnect imperfections, surface roughness, and thermal mismatch between different materials [12,205]. These effects are particularly critical in antenna-in-package and antenna-on-package implementations, where the antenna, feeding network, RF front-end, and packaging environment must be co-designed to reduce parasitic coupling, insertion losses, and thermal stress [204,205].

Complex metamaterial and metasurface-based antennas may also require dense periodic patterns, suspended layers, or precise air gaps, which can increase fabrication complexity and reduce repeatability in practical manufacturing [12,140]. Therefore, although these structures are promising for high-performance compact MIMO antennas, their practical deployment requires fabrication-tolerant geometries, low-loss substrates, robust feeding transitions, thermal-aware design, and experimental validation under realistic manufacturing conditions [12,140,204].

5.5. Summary of Key Design Trade-Offs

Designing compact mmWave antennas for 5G requires balancing several critical trade-offs. Miniaturizing antenna elements improves compactness but can increase mutual coupling, reducing diversity and link reliability; this can be mitigated using techniques such as DGS, EBG, or neutralization lines [206]. High-gain structures enhance link efficiency and help overcome severe path losses, but they typically narrow the operational bandwidth, whereas broadband designs provide spectral flexibility at the cost of lower gain and efficiency [12]. Similarly, achieving high performance often requires low-loss substrates and precise fabrication methods, while cost-effective materials like FR-4 enable practical, large-scale production [207]. Carefully balancing compactness, isolation, bandwidth, gain, performance, and manufacturability is therefore essential to develop reliable, efficient, and economically viable 5G mmWave antennas.

5.6. Integration and Manufacturing Challenges

Compact MIMO antennas for 5G mmWave systems face significant integration and manufacturing challenges due to the extremely small physical dimensions of radiating elements, the high sensitivity to fabrication tolerances, and the elevated cost of suitable high-frequency materials. At mmWave frequencies, miniaturization greatly increases mutual coupling and makes antenna performance highly dependent on substrate variations, connector effects, and small mechanical inaccuracies, which complicates reliable mass production [140]. High-performance low-loss materials—such as PTFE, LCP, RO5880, or microwave ceramics—require advanced fabrication processes with tighter tolerances, while copper surface roughness, dimensional deviations, and realistic feed-connector designs can significantly degrade gain, bandwidth, and efficiency [208]. Furthermore, the integration of antennas within compact modules or antenna-in-package (AiP) platforms demands careful co-design of the antenna, RFIC, and packaging, along with addressing thermal management and power-consumption constraints inherent to highly integrated mmWave

MIMO architectures [12]. These technical issues are compounded by industrial constraints, where balancing performance, cost, and manufacturability remains a major obstacle for scalable production of compact mmWave MIMO systems [209].

6. Trends and Future Research Directions

Although several compact mmWave MIMO antenna systems have already provided interesting performance in terms of isolation, bandwidth, gain, and ECC, multiple open challenges remain before their extensive incorporation into forthcoming 5G-advanced and 6G systems. Future research is expected to move toward more adaptive, intelligent, compact, energy-efficient, and sustainable antenna architectures.

This section expands the "Trends and future research directions" branch of Figure 2 by discussing reconfigurable antennas, 6G/sub-THz integration, self-powered and eco-friendly antennas, and IoT/V2X-oriented antenna systems.

6.1. Reconfigurable and Smart Antennas

Reconfigurable and smart antennas are becoming essential for future 5G-advanced and 6G mmWave MIMO systems, as they enable dynamic control of key antenna characteristics such as operating frequency, polarization, radiation pattern, and beam direction. This adaptability is particularly important at mmWave frequencies, where high propagation losses, blockage effects, mutual coupling, and miniaturization constraints can strongly degrade system performance. Unlike conventional fixed antennas, reconfigurable antennas can adapt their electromagnetic behavior in real time according to the propagation environment and communication requirements [210,211].

The practical implementation of these antennas relies mainly on the integration of tunable components and smart materials, including PIN diodes, RF-MEMS, varactors, liquid crystals, and graphene-based materials. PIN diodes and varactors are widely used for frequency and pattern reconfiguration due to their compactness and simple biasing mechanisms, while RF-MEMS offer low insertion loss, high linearity, and good isolation, making them attractive for high-performance reconfigurable RF front-ends [210,212,213]. In addition, liquid crystals and graphene-based materials provide promising solutions for continuous tunability and compact integration, especially for mmWave and future THz antenna systems [211,214].

In compact MIMO configurations, reconfigurable and smart antennas can improve isolation, reduce the envelope correlation coefficient (ECC), enhance beam steering, and support adaptive polarization diversity. For example, polarization-reconfigurable MIMO antennas can achieve isolation above 30 dB [6], while metasurface-based reconfigurable designs can provide frequency and radiation-pattern switching with isolation up to -40 dB [215]. Therefore, future research should focus on low-loss, low-power, and fabrication-compatible smart materials and tuning mechanisms in order to improve the reliability, miniaturization, and practical integration of compact mmWave MIMO antennas [211,216,217].

Beyond frequency and polarization reconfigurability, future 6G antenna systems will require adaptive radiation control and beam-steering capability to overcome blockage, user mobility, and highly directional propagation. Recent beam-scanning architectures show that reconfigurable transmission-line and metamaterial-inspired concepts can dynamically control the radiation direction without relying only on fixed-frequency operation. For example, Wang et al. [218] proposed an orthogonal quad-beam scanning antenna based on 1-bit dielectric modulation in a plasmonic metamaterial transmission line, achieving continuous quad-beam scanning over more than 60° across two orthogonal planes. Although this design is not a compact mmWave MIMO antenna, it highlights an important

future direction where compact MIMO systems should evolve toward beam-steerable, reconfigurable, and environment-adaptive radiation architectures.

6.2. Integration with 6G and Beyond Technologies

Anticipating 6G and further advancements, mmWave MIMO systems are projected to progress towards sub-THz and THz frequency bands to facilitate very high data speeds, minimal latency, extensive connectivity, and improved spatial resolution. The large available bandwidth at THz frequencies makes this spectrum a promising candidate for future wireless systems, especially for short-range ultra-high-speed links, high-resolution sensing, and Integrated Sensing and Communication (ISAC) applications [219,220]. In this context, improved MIMO architectures, such as massive MIMO, sparse MIMO, cell-free MIMO, and distributed MIMO, will be pivotal in enhancing beamforming performance, spectral efficiency, and coverage in highly directed propagation situations [221,222].

For future 6G systems, holographic metasurfaces and extremely large-scale MIMO are emerging as promising technologies for achieving highly integrated, energy-efficient, and programmable antenna platforms. Di et al. [223] demonstrated the relevance of holographic metasurfaces for extremely large-scale MIMO communications, highlighting their potential for compact smart antenna integration, programmable wavefront control, and next-generation mmWave/6G wireless systems.

Beyond terrestrial networks, future 6G systems are also expected to integrate non-terrestrial and extreme-environment communication scenarios. Space-based networks, including satellite constellations, high-altitude platforms, UAV-assisted networks, and interplanetary communication links, will require compact, high-gain, and highly directional antenna systems capable of maintaining reliable connectivity over long distances. In such scenarios, mmWave and THz MIMO antennas can support narrow-beam transmission, high-capacity inter-satellite links, joint communication and positioning, and space-based sensing applications [224,225].

Underwater communications are another emerging direction for 6G-enabled global connectivity. However, unlike space and terrestrial scenarios, direct mmWave and THz transmission is strongly limited underwater due to severe absorption and propagation losses in aquatic environments. Therefore, future underwater 6G systems will likely rely on hybrid communication architectures combining acoustic, optical, magnetic-induction, and RF-based links, depending on the required range, data rate, latency, and energy consumption. In this context, compact and adaptive antenna systems may contribute mainly to surface gateways, buoy-to-satellite links, autonomous underwater vehicle coordination, and the connection between underwater sensor networks and terrestrial or non-terrestrial 6G infrastructures [226,227].

Despite their strong potential, the integration of mmWave, THz, space, and underwater communication technologies introduces several open challenges. At THz frequencies, severe path loss, molecular absorption, hardware losses, thermal effects, and strict fabrication tolerances can significantly affect antenna performance. Moreover, the practical deployment of compact MIMO antennas for 6G requires accurate fabrication, advanced packaging, low-loss substrates, efficient beamforming circuits, and reliable integration with RF front-end modules. Emerging fabrication methods such as 3D-printed metallic structures, micromachining, and additive manufacturing may partially address these challenges by enabling complex, lightweight, and compact antenna geometries suitable for future 6G and beyond systems [219,224,228].

From a practical deployment perspective, the most realistic antenna technologies for sub-THz 6G systems are expected to be low-loss antenna-in-package (AiP) and antenna-on-package (AoP) arrays, substrate-integrated waveguide (SIW)-based antennas, dielectric

resonator antennas (DRAs), micromachined metallic waveguide antennas, and compact phased-array architectures [71,204,205,229]. These technologies offer a better compromise between radiation efficiency, feeding network losses, fabrication accuracy, packaging compatibility, and thermal stability. In particular, AiP/AoP solutions are attractive for compact integration with RF front-end modules, while SIW-based and waveguide-based antennas can reduce feeding losses at high frequencies. DRAs are also promising because they avoid conductor losses in the radiating element and can provide high radiation efficiency at mmWave and sub-THz frequencies [71]. In contrast, fully on-chip antennas, complex multilayer metamaterial structures, and highly dense reconfigurable metasurfaces remain promising but less mature for large-scale deployment because they may suffer from high conductor and dielectric losses, thermal issues, biasing complexity, and strong sensitivity to fabrication tolerances at sub-THz frequencies [204,229]. Therefore, near-term realistic sub-THz 6G antenna development should prioritize low-loss materials, short and efficient feeding paths, packaging-aware co-design, thermal management, and fabrication-tolerant array architectures.

6.3. Self-Powered and Eco-Friendly Antennas

Eco-friendly and self-powered antennas are emerging as a major research direction. Metasurface energy harvesters combine RF energy harvesting with reconfigurable functionality for self-sustaining intelligent surfaces [230], while flexible mmWave rectennas enable efficient RF–DC conversion in wearable and portable devices [231]. Biodegradable materials such as chitosan [232] and 3D-printed polymers including BioFila and graphene [233] provide sustainable alternatives, and hybrid MIMO architectures integrating RF harvesting with perovskite photovoltaics achieve high efficiency for compact IoT applications [234].

6.4. Antennas for IoT and V2X

For IoT and V2X systems, strict size, integration, and cost constraints drive the development of highly compact MIMO mmWave antennas. Designs targeting sub-20 mm $\times$ 20 mm formats achieve gains above 7 dBi and efficiencies over 75% through optimized geometries and practical substrates [235]. T-shaped radiators, CSRR-loaded grounds, DGS, slots, stubs, DRAs, and metamaterials enable wideband, high-gain, and low-coupling performance without expensive fabrication, supporting cost-sensitive 5G NR IoT and windshield-mounted V2X antennas [11,236,237].

Recent advancements emphasize the synergy of antenna hardware and advanced signal-processing techniques. On-chip beamforming integration allows real-time steering up to $\pm 45^{\circ}$ and adaptive switching between radar and 5G V2X communication tasks [238]. Hybrid analog-digital architectures balance performance, complexity, cost, and energy efficiency [239]. Artificial intelligence—including DNNs, CNNs, and reinforcement learning—accelerates EM simulations, optimizes beam-steering arrays, and enables intelligent real-time adaptation of radiation patterns, improving S-parameters, gain, efficiency, and overall system performance [240,241].

In addition to electromagnetic performance metrics, the practical deployment readiness of the reviewed technologies should also be considered. In this review, maturity is assessed according to the reported validation level, ranging from simulation-only studies, measured laboratory prototypes, packaging-aware or system-level prototypes, to technologies that are closer to commercial integration. Most compact mmWave MIMO antenna designs reported in the academic literature remain at the simulation or laboratory-prototype stage, where validation is mainly based on S-parameters, radiation patterns, isolation, ECC, CCL, gain, and efficiency. Therefore, high antenna-level performance does not necessarily imply immediate readiness for real 5G products or future 6G deployment.

To further clarify the practical maturity, deployment readiness, and expected impact of these future research directions. The discussed technologies can be broadly classified into three levels. First, low-loss AiP/AoP solutions, SIW-based antennas, dielectric resonator antennas, compact phased arrays, and IoT/V2X-oriented mmWave MIMO antennas represent relatively mature and near-term deployable directions, since they are more compatible with existing RF front-end integration, packaging processes, and practical fabrication constraints [71,235,236]. Second, reconfigurable antennas, metasurface-assisted MIMO structures, and AI-assisted antenna optimization are highly promising medium-term directions, offering strong potential for adaptive beam control, isolation enhancement, design acceleration, and improved spectral efficiency, but they still require further experimental validation, reliable biasing networks, and robust system-level integration [165,166]. Finally, THz/sub-THz antenna systems, holographic metasurfaces, self-powered antennas, and biodegradable or eco-friendly antenna materials are expected to have a high long-term impact on 6G and beyond networks, particularly for ultra-high-speed links, integrated sensing and communication, sustainable IoT, and intelligent radio environments [232,233]. However, these technologies remain less mature for large-scale deployment due to challenges related to propagation loss, fabrication tolerance, material reliability, thermal effects, and integration complexity [204,229].

Together, these trends indicate that future 5G/6G mmWave MIMO systems will combine reconfigurable hardware, advanced MIMO architectures, AI-assisted optimization, self-powered operation, and eco-friendly materials to achieve high performance, miniaturization, and sustainability for IoT, V2X, and space-based applications.

Despite the progress achieved in compact mmWave MIMO antennas, several unresolved issues still limit their transition from reported prototypes to reliable 5G-advanced and 6G deployments. The main difficulty is not only to improve one performance metric, but to maintain a balanced compromise between compactness, wide bandwidth, high gain, high isolation, low ECC, low CCL, radiation efficiency, and fabrication feasibility within the same antenna platform [12,140]. Many reported designs achieve excellent isolation or gain, but this improvement is often obtained through additional slots, DGS, metamaterial cells, multilayer structures, or complex feeding networks, which may increase fabrication sensitivity and reduce scalability [13,140].

A major technical bottleneck is the scalable integration of dense multi-port MIMO antennas. Moving from two-port or four-port designs toward eight-port or higher-order arrays increases mutual coupling, feeding-network complexity, phase imbalance, and sensitivity to manufacturing errors [12,13]. This issue becomes more critical at sub-THz frequencies, where conductor losses, dielectric losses, packaging parasitics, thermal effects, and strict fabrication tolerances strongly affect the measured performance [204,205]. Therefore, future designs should not only be evaluated through simulated antenna metrics, but also through experimentally validated packaging-aware, thermal-aware, and fabrication-tolerant prototypes.

Another open problem concerns the practical maturity of advanced antenna concepts such as metasurfaces, metamaterial-loaded structures, multilayer antennas, and reconfigurable arrays. These technologies are promising for compactness, isolation enhancement, and adaptive radiation control. However, their deployment still requires reliable biasing networks, low-loss materials, repeatable manufacturing processes, and robust RF front-end integration [140,204]. In parallel, AI-assisted antenna optimization remains promising for reducing simulation time and exploring large design spaces, but its reliability depends on training-data quality, physical interpretability, and experimental validation [165,166]. These unresolved issues represent key directions for future compact mmWave and sub-THz MIMO antenna research.

7. Conclusions

This review presents a comprehensive analysis of compact, multi-port millimeter wave (mmWave) Multiple-Input–Multiple-Output (MIMO) antenna systems for fifth-generation (5G) and beyond wireless communications. The study highlighted major advances in antenna structure, including microstrip patch, slot-based, dielectric resonator, and metamaterial-inspired antennas, as well as array-based configurations. These designs demonstrate strong potential for achieving a wide bandwidth, high gain, a low envelope correlation coefficient (ECC), a high diversity gain (DG), reduced channel capacity loss (CCL), and stable multi-port operation.

A key finding of this review is that compactness and electromagnetic performance are strongly interdependent; while reducing the size of the antenna facilitates its integration into mobile terminals, IoT devices, V2X platforms, and antenna-in-package modules, it also increases mutual coupling between closely spaced radiating elements. Several coupling reduction techniques have been investigated to address this issue, including defected ground structures (DGSs), electromagnetic bandgap structures (EBG), parasitic elements, neutralization lines, polarization diversity, spatial diversity, metamaterials, and CSRR-based approaches; while these techniques have shown clear improvements in isolation and diversity performance, they often introduce additional design complexity, fabrication sensitivity, or bandwidth limitations.

The review emphasized the importance of substrate and material technologies in millimeter wave (mmWave) Multiple-Input–Multiple-Output (MIMO) antenna design. Low-loss substrates, such as Rogers RT/Duroid materials, are widely used due to their stable dielectric behavior and high radiation efficiency. Meanwhile, flexible substrates, surface-mount integrated waveguide (SIW)-based platforms, multilayer structures, 3D-printed materials, and metamaterial-based substrates offer new possibilities for wearable, conformal, compact, and highly integrated antenna systems. However, when choosing a substrate, it is important to balance dielectric performance, cost, manufacturability, thermal stability, and tolerance sensitivity.

The importance of researching compact millimeter wave (mmWave) Multiple-Input–Multiple-Output (MIMO) antennas lies in its direct contribution to developing future wireless communication systems. As 5G-advanced and 6G networks transition to higher frequencies, greater bandwidths, massive connectivity, and ultra-low latency, antenna systems will be crucial for establishing reliable, high-capacity connections. These antennas enable spatial multiplexing, beamforming, diversity enhancement, and improved spectral efficiency. These capabilities are essential for overcoming severe propagation losses and blockage effects associated with mmWave.

This research is important because it connects antenna design with broader, system-level requirements. Future wireless networks will rely not only on high-performance radiating elements but also on intelligent, adaptive, energy-efficient, and manufacturable antenna architectures. Key enablers for next-generation communication systems are expected to include AI-assisted optimization, reconfigurable antennas, smart materials, RIS-based environments, antenna-in-package integration, and sustainable fabrication methods. Progress in compact millimeter wave Multiple-Input–Multiple-Output (MIMO) antennas will support traditional mobile communications as well as emerging applications such as the Internet of Things (IoT), vehicle-to-everything (V2X) communication, integrated sensing and communication, satellite networks, wearable devices, and smart environments.

Overall, the evolution of compact millimeter wave (mmWave) Multiple-Input–Multiple-Output (MIMO) antennas is moving toward more intelligent, adaptive, and application-oriented designs. Future antenna systems must combine high gain, wide bandwidth, high isolation, low correlation, low loss, and compact size, all while remaining

compatible with practical fabrication and large-scale integration. Simultaneously achieving all these requirements remains a major challenge, especially at sub-THz and THz frequencies, where material losses, manufacturing tolerances, thermal effects, and packaging constraints are more critical.

Future research should focus on hybrid design strategies combining advanced geometries, low-loss substrates, metamaterials, reconfigurable components, AI-based optimization, and sustainable manufacturing technologies. Particular attention should be given to scalable, multi-port architectures; fabrication-tolerant designs; energy-efficient beamforming; and reliable integration with RF front-end circuits. In this context, compact millimeter wave Multiple-Input–Multiple-Output (MIMO) antennas will continue to be a fundamental technology for 5G-advanced, 6G, and future wireless communication systems.

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